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Sharp Poincaré interpolation along Wasserstein geodesics

Bang-Xian Han* Zhuo-Nan Zhu†

Abstract

Let μ_0 and μ_1 be κ_0- and κ_1-strongly log-concave probability measures on $\mathbb{R}^n$, and let $(\mu_t)_{t\in[0,1]}$ be their quadratic Wasserstein geodesic. We prove the sharp Poincaré constant estimate

$$\sqrt{C_P(\mu_t)} \le \frac{1-t}{\sqrt{\kappa_0}} + \frac{t}{\sqrt{\kappa_1}}.$$

The coefficient is optimal for every t, κ_0, κ_1, and the result remains valid for extended-valued potentials without symmetry assumptions. Equality at an interior time holds exactly when the two endpoints have Gaussian factors in the same direction, with variances κ_0^{-1} and κ_1^{-1}. All such directions form a maximal linear subspace. This gives an affirmative answer, without symmetry or parity assumptions, to a question of Aishwarya–Rotem.

The proof develops a coupled Bochner method for the two endpoints. We solve a weighted Poisson equation involving the Hessian of the Brenier potential. From its solution we construct a Bochner couple, and separate estimates for its two fields are combined along the interpolation. No curvature bound is needed for the intermediate measures. Applications include centered Gaussian relative entropy, Gaussian Brunn–Minkowski inequalities with barycenter terms, and centered HWI, logarithmic Sobolev, and Talagrand estimates.

Keywords: Poincaré inequality, coupled Bochner method, log-concavity, optimal transport, Wasserstein geodesic, relative entropy, Gaussian Brunn–Minkowski inequality

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*School of Mathematics, Shandong University, Jinan 250100, China. Email: hanbx@sdu.edu.cn

†School of Mathematical Sciences, University of Science and Technology of China, Hefei 230026, China. Email: zhuonanzhu@mail.ustc.edu.cn

1 Introduction

1.1 Background and main results

We study whether analytic estimates implied by curvature bounds at two endpoints remain valid along their quadratic Wasserstein geodesic. Convexity of the intermediate measures cannot be assumed: even for uniform measures on convex bodies, the support of a Wasserstein midpoint can be nonconvex [45], and the midpoint is then not log-concave.

Optimal transport has long linked interpolation inequalities, entropy, and curvature. Otto and Villani proved, under curvature hypotheses, the implication from logarithmic Sobolev to Talagrand inequalities and the HWI estimate [44]. Cordero-Erausquin, McCann, and Schmuckenschläger established a Riemannian interpolation inequality through optimal transport [13]. Von Renesse and Sturm characterized lower Ricci curvature by displacement convexity of entropy and equivalent transport and gradient estimates [49]. On Riemannian manifolds with boundary, the first author subsequently proved measure-rigidity results for the corresponding curvature-dimension conditions [26]. These results motivate the study of spectral inequalities along a prescribed Wasserstein geodesic when curvature assumptions are imposed only at its endpoints.

We prove that positive curvature bounds at the endpoints yield a Poincaré estimate along their quadratic Wasserstein geodesic. The sharp estimate is linear in the scale $\sqrt{C_P(\mu)}$, which is the standard deviation for an isotropic Gaussian. If the endpoint (Bakry–Émery) curvature bounds are κ_0 and κ_1, then

$$\sqrt{C_P(\mu_t)} \le (1-t)\kappa_0^{-1/2} + t\kappa_1^{-1/2}.$$

Gaussian endpoints attain equality, so the coefficient is optimal. The argument does not require a lower Hessian bound for the intermediate potentials.

The proof links the two endpoint Bochner inequalities through one weighted Poisson equation. It does not propagate convexity of the potentials along the geodesic. The source and target estimates are proved separately. This keeps their sharp constants and is also important for the equality case. We call this argument the *coupled Bochner method*.

The basic object is a *Bochner couple*, a pair of vector fields at the two endpoints. The fields have the same divergence datum and are related by $DT = \operatorname{Hess}\Phi$, the differential of the Brenier map. Each field satisfies its own Bochner estimate. We first construct this pair from a single scalar Poisson solution. We then interpolate the two fields to obtain the estimate at time t. We refer to this construction and interpolation as *Bochner coupling*.

For a probability measure $\mu \in \mathcal{P}(\mathbb{R}^n)$, its Poincaré constant $C_P(\mu)$ is defined as the least $C \in [0,\infty]$ such that

$$\operatorname{Var}_\mu(f) := \int \left| f - \int f \,\mathrm{d}\mu \right|^2 \mathrm{d}\mu \le C \int |\nabla f|^2 \,\mathrm{d}\mu, \qquad \text{for every } f \in C_c^\infty(\mathbb{R}^n).$$

For integrable functions f and g, we use

$$\operatorname{Cov}_\mu(f,g) := \int \left(f - \int f\,\mathrm{d}\mu\right)\left(g - \int g\,\mathrm{d}\mu\right) \mathrm{d}\mu.$$

If μ has a finite first moment, write its barycenter $b_\mu := \int x \,\mathrm{d}\mu$. If μ has a finite second moment, its covariance matrix is

$$\operatorname{Cov}(\mu) := \int (x - b_\mu) \otimes (x - b_\mu) \,\mathrm{d}\mu.$$

For a random vector X with law μ, we write $\mathrm{Cov}(X) := \mathrm{Cov}(\mu)$.

We write $\mathrm{d}\mu = e^{-V}\,\mathrm{d}x$ for a probability measure, absorbing the normalizing constant into the potential V. We assume throughout that $V : \mathbb{R}^n \to (-\infty, +\infty]$ is proper and lower semicontinuous. We call μ log-concave if V is convex. For $\kappa > 0$, we call μ κ-strongly log-concave if $V - \kappa\frac{|x|^2}{2}$ is convex. In the latter case, the classical Brascamp–Lieb inequality [6] yields $C_P(\mu) \le \kappa^{-1}$.

Now let μ_0 and μ_1 be strongly log-concave. Since μ_0 is absolutely continuous, Brenier's theorem [8] provides an optimal transport map T from μ_0 to μ_1. The constant-speed W_2-geodesic joining μ_0 and μ_1 is given by

$$F_t = (1-t)I + tT, \qquad \mu_t = (F_t)_{\#}\mu_0, \qquad t \in [0,1].$$

Here W_2 denotes the quadratic Wasserstein distance and I is the identity map. The strong log-concavity assumptions give Brascamp–Lieb bounds at the two endpoints, but no corresponding lower Hessian bound for the potential of μ_t is available a priori. Thus the Brascamp–Lieb inequality cannot be applied directly at intermediate times.

The following theorem gives the sharp bound for possibly different endpoint curvatures, without any symmetry assumption. For $i = 0, 1$, let μ_i be κ_i-strongly log-concave for some $\kappa_i > 0$. We allow the corresponding potentials to take the value $+\infty$; in particular, the result includes Gaussian measures conditioned on full-dimensional convex sets.

Theorem 1.1 (Sharp Poincaré interpolation)**.** *For every $t \in [0,1]$, we have*

$$\mathrm{Var}_{\mu_t}(f) \le \left(\frac{1-t}{\sqrt{\kappa_0}} + \frac{t}{\sqrt{\kappa_1}}\right)^2 \int |\nabla f|^2\,\mathrm{d}\mu_t, \qquad \text{for every } f \in C_c^\infty(\mathbb{R}^n). \tag{1.1}$$

Equivalently,

$$\sqrt{C_P(\mu_t)} \le \frac{1-t}{\sqrt{\kappa_0}} + \frac{t}{\sqrt{\kappa_1}}.$$

Remark 1.2. The coefficient in (1.1) is optimal for every $t \in [0,1]$ and every $\kappa_0, \kappa_1 > 0$. Indeed, let $\mu_i = N(0, \kappa_i^{-1} I)$ for $i = 0, 1$, where $N(m, \Sigma)$ denotes the Gaussian law with mean m and covariance matrix Σ. Then $T(x) = \sqrt{\kappa_0/\kappa_1}\,x$ is the unique optimal transport map from μ_0 to μ_1. It follows that

$$\mu_t = N\left(0, \left(\frac{1-t}{\sqrt{\kappa_0}} + \frac{t}{\sqrt{\kappa_1}}\right)^2 I\right).$$

Thus the coefficient is optimal because $C_P(N(0, a^2 I)) = a^2$.

Remark 1.3. A stronger statement would replace $\kappa_i^{-1/2}$ in Theorem 1.1 by $\sqrt{C_P(\mu_i)}$; that is, it would assert

$$\sqrt{C_P(\mu_t)} \le (1-t)\sqrt{C_P(\mu_0)} + t\sqrt{C_P(\mu_1)}.$$

This interpolation bound in terms of the endpoint Poincaré constants fails in general, even for strongly log-concave endpoints; see Proposition A.1.

We next characterize equality at an interior time.

Theorem 1.4 (Equality characterization)**.** *Fix $t \in (0,1)$ and denote $m_i = \int x\,\mathrm{d}\mu_i$. Then*

$$C_P(\mu_t) = \left(\frac{1-t}{\sqrt{\kappa_0}} + \frac{t}{\sqrt{\kappa_1}}\right)^2$$

if and only if there exists a unit vector $e \in \mathbb{R}^n$ *such that, under the orthogonal decomposition* $\mathbb{R}^n = \mathbb{R}e \oplus e^\perp$*, the coordinate map* $\mathcal{C}_e(x) = (e \cdot x, P_{e^\perp} x)$ *satisfies*

$$(\mathcal{C}_e)_{\#}\mu_i = N\big(\langle e, m_i\rangle, \kappa_i^{-1}\big) \otimes \nu_i, \qquad \nu_i \in \mathcal{P}(e^\perp) \qquad i = 0, 1. \tag{1.2}$$

In this case the optimal transport splits along the same decomposition and

$$C_P(\mu_s) = \left(\frac{1-s}{\sqrt{\kappa_0}} + \frac{s}{\sqrt{\kappa_1}}\right)^2, \qquad \text{for every } s \in [0,1].$$

The restriction $t \in (0,1)$ is essential in the rigidity statement. At an endpoint, equality constrains only that endpoint and cannot force a Gaussian factor in the other measure. The proof also identifies the maximal subspace of directions along which both endpoints split off the corresponding Gaussian factors; see Proposition 4.2. In [3, Question 1.6], Aishwarya and Rotem asked about the unit Poincaré bound along the optimal Wasserstein interpolation between even 1-strongly log-concave endpoints. Their question was restricted to odd test functions; see also [2, Question 1.4]. Theorem 1.1 proves a stronger result. It requires neither symmetry of the endpoints nor a parity assumption on the test function. Aishwarya and Rotem proved their formulation in dimension one [3, Theorem 4.11], for two Gaussian endpoints, and when one endpoint is the standard Gaussian [3, Proposition 4.12].

Aishwarya and Li used the common-source coupling $(T_0(Z), T_1(Z))$, where Z is standard Gaussian and T_i is the Brenier map from the Gaussian measure to the i-th endpoint. Every interpolant $((1-t)T_0 + tT_1)(Z)$ is a 1-Lipschitz image of Z and therefore satisfies the full Poincaré inequality with constant 1 [2, Remark 1]. This coupling is generally not optimal between the two prescribed endpoints. To the best of our knowledge, the full Poincaré bound for the quadratic optimal coupling of the prescribed endpoints had remained open. Theorem 1.1 proves it along the quadratic Wasserstein geodesic between the prescribed endpoints.

The full Poincaré estimate also controls the fluctuation of the velocity gradient in the second variation of Gaussian relative entropy. We first remove the translation component determined by the means and then apply the estimate componentwise. Let γ be the standard Gaussian measure on $\mathbb{R}^n$. For a probability measure μ, define its relative entropy with respect to γ by

$$\mathrm{Ent}_\gamma(\mu) = \int \rho \log \rho \, \mathrm{d}\gamma \quad \text{if } \mathrm{d}\mu = \rho \, \mathrm{d}\gamma,$$

with the convention $0 \log 0 = 0$, and set $\mathrm{Ent}_\gamma(\mu) = +\infty$ otherwise. Define

$$\mathcal{D}(\mu) = \mathrm{Ent}_\gamma(\mu) - \frac{1}{2}|b_\mu|^2.$$

Thus $\mathcal{D}(\mu)$ is the *centered Gaussian relative entropy* of μ; equivalently, it is the Gaussian relative entropy of the centered translate of μ. For two measures, define

$$W_{\mathrm{c}}^2 = W_2^2(\mu_0, \mu_1) - |b_{\mu_1} - b_{\mu_0}|^2.$$

Theorem 1.5 (Centered Gaussian relative entropy)**.** *Let* μ_0 *and* μ_1 *be* 1*-strongly log-concave, and let* $(\mu_t)_{0 \le t \le 1}$ *be their quadratic Wasserstein geodesic. Then* $0 \le W_{\mathrm{c}}^2 \le 2n$*, and* $\mathcal{D}_t = \mathcal{D}(\mu_t)$ *satisfies the following distortion inequality along the whole geodesic:*

$$e^{-\mathcal{D}_t/n} \ge \frac{\sin((1-t)\omega)}{\sin\omega} e^{-\mathcal{D}_0/n} + \frac{\sin(t\omega)}{\sin\omega} e^{-\mathcal{D}_1/n}, \qquad \omega = \sqrt{\frac{2}{n}}\, W_{\mathrm{c}}. \tag{1.3}$$

The sine quotients are understood by continuity when $\omega = 0$, in which case they equal $1-t$ and t, respectively. Moreover, equality in (1.3) at some $t \in (0,1)$ holds if and only if μ_1 is a translate of μ_0. In this case, equality holds for every $t \in [0,1]$.

In particular, if the endpoints have the same barycenter, then the same statements hold with $\mathrm{Ent}_\gamma(\mu_t)$ in place of $\mathcal{D}_t$ and with $W_2(\mu_0,\mu_1)$ in place of W_c. In this case, equality at an interior time holds if and only if $\mu_0 = \mu_1$.

Infinitesimal form. Under the smooth two-sided Hessian hypotheses of Section 2, the proof first establishes

$$\mathcal{D}_t'' \geq 2W_c^2 + \frac{1}{n}(\mathcal{D}_t')^2. \tag{1.4}$$

This is the differential form of the $(2,n)$-convexity estimate; the finite-time inequality (1.3) follows by a one-dimensional comparison argument and then by regularization.

When the endpoints have the same barycenter, Theorem 1.5 establishes $(2,n)$-convexity along the Wasserstein geodesic for arbitrary 1-strongly log-concave measures. Aishwarya and Rotem obtained the same differential estimate under a pathwise odd Poincaré assumption [3, Theorem 4.6] and verified this assumption in dimension one and along two further classes of geodesics [3, Theorem 4.11 and Proposition 4.12]. For even endpoints, Aishwarya and Li obtained an entropy inequality with sine coefficients along the interpolation of the two Brenier maps transporting the standard Gaussian measure to the respective endpoints. They also characterized equality in the resulting affine inequality [2, Theorem 1.5 and Remark 1]. The present theorem requires neither symmetry nor a pathwise odd Poincaré assumption. It proves the inequality with sine coefficients along the quadratic Wasserstein geodesic and characterizes equality.

Theorem 1.5 also yields a Gaussian Brunn–Minkowski inequality. Gardner and Zvavitch conjectured the dimensional Gaussian Brunn–Minkowski inequality for convex bodies containing the origin [25]. Nayar and Tkocz disproved this conjecture in the nonsymmetric case, even under the assumption that both bodies contain the origin, and proposed its origin-symmetric form [43]. Following important local work of Kolesnikov and Livshyts [37], the origin-symmetric conjecture was proved by Eskenazis and Moschidis [22]. Corollary 1.6 replaces origin symmetry by the assumption that the conditioned Gaussian measures have the same barycenter. Without this assumption, it gives an inequality with explicit barycenter terms. For a convex Borel set K with $\gamma(K) > 0$, denote

$$\gamma_K(A) = \frac{\gamma(A \cap K)}{\gamma(K)} \qquad \text{and} \qquad b_K = \int x \, \mathrm{d}\gamma_K(x),$$

the Gaussian measure conditioned on K and its barycenter, respectively.

Corollary 1.6 (Gaussian Brunn–Minkowski with barycenter terms)**.** *Let $K_0, K_1 \subset \mathbb{R}^n$ be convex Borel sets with $\gamma(K_i) > 0$. For every $t \in [0,1]$, set $K_t = (1-t)K_0 + tK_1$ and $b_t = (1-t)b_{K_0} + tb_{K_1}$. Then*

$$e^{|b_t|^2/(2n)}\gamma(K_t)^{1/n} \geq (1-t)e^{|b_{K_0}|^2/(2n)}\gamma(K_0)^{1/n} + te^{|b_{K_1}|^2/(2n)}\gamma(K_1)^{1/n}. \tag{1.5}$$

Moreover, equality in (1.5) at some $t \in (0,1)$ holds if and only if $\bar{K}_0 = \bar{K}_1$. In this case, equality holds for every $t \in [0,1]$.

In particular, if $b_{K_0} = b_{K_1}$, then

$$\gamma(K_t)^{1/n} \geq (1-t)\gamma(K_0)^{1/n} + t\gamma(K_1)^{1/n}.$$

The condition $b_{K_0} = b_{K_1}$ does not require symmetry. For example, a regular simplex centered at the origin is invariant under its full symmetry group. It follows that the Gaussian measure conditioned on the simplex has barycenter zero. The simplex is not centrally symmetric when $n \geq 2$. Section 5 gives a functional version. It extends the Gaussian Borell–Brascamp–Lieb inequality of Aishwarya and Li beyond even log-concave functions. The new assumption is that the probability measures proportional to $f\,\mathrm{d}\gamma$ and $g\,\mathrm{d}\gamma$ have the same barycenter. We also provide HWI, logarithmic Sobolev, Talagrand, and Ornstein–Uhlenbeck estimates after centering.

1.2 Proof strategy: coupled endpoint Bochner estimates

The Bochner identity is used only at μ_0 and μ_1. We construct one vector field at each endpoint and interpolate these fields. Thus no curvature estimate is required for the potential of the intermediate measure. Treating the endpoints separately gives the optimal coefficient and makes the equality case accessible.

Step 1: the weighted Poisson equation. We first work in the smooth setting. Let $T = \nabla\Phi$ be the Brenier map and set $H = \operatorname{Hess}\Phi$. For a centered function $g \in L^2(\mu_0)$, we solve

$$-\operatorname{div}_{\mu_0}(H^{-1}\nabla u) = g$$

and define

$$X = H^{-1}\nabla u, \qquad Y \circ T = \nabla u.$$

The same scalar function u defines both fields, while H relates them through $Y \circ T = HX$. Smooth approximations of (X, Y) are Bochner couples in the sense of Definition 2.3. A change-of-variables identity for weighted divergence shows that the two fields have the same divergence datum.

Step 2: Bochner estimates at the two endpoints. At the source, differentiating $HX = \nabla u$ and using the symmetry of the third derivatives of Φ shows that

$$H\,DX \quad \text{is symmetric.}$$

At the target, the chain rule gives

$$((DY) \circ T)H = \operatorname{Hess} u,$$

so this matrix is also symmetric. Because H is positive definite, each identity makes the corresponding trace term in the vector Bochner formula nonnegative. Applying that formula at the two endpoints gives

$$\|X\|_{L^2(\mu_0)} \leq \kappa_0^{-1/2}\,\|g\|_{L^2(\mu_0)}\,, \qquad \|Y\|_{L^2(\mu_1)} \leq \kappa_1^{-1/2}\,\|g\|_{L^2(\mu_0)}\,.$$

Both estimates are sharp. We keep them separate until the final interpolation step.

Step 3: interpolation of the two fields. Let $F_t = (1-t)I + tT$ and define q_t on μ_t by

$$q_t \circ F_t = (1-t)X + t(Y \circ T).$$

The same change-of-variables formula for weighted divergence yields

$$-(\operatorname{div}_{\mu_t} q_t) \circ F_t = g.$$

The separate endpoint estimates imply

$$\|q_t\|_{L^2(\mu_t)} \le \left(\frac{1-t}{\sqrt{\kappa_0}} + \frac{t}{\sqrt{\kappa_1}}\right) \|g\|_{L^2(\mu_0)} .$$

Taking $g = (f - \int f \,\mathrm{d}\mu_t) \circ F_t$ and integrating by parts proves (1.1) in the smooth setting. The general case follows by regularization and stability of optimal transport.

This interpolation of the Bochner couple is the coupling step. The endpoint norms combine linearly before the final Cauchy–Schwarz inequality. This gives the sharp coefficient.

For the equality case, take a sequence that approaches equality in the Poincaré estimate. The two endpoint estimates and the triangle inequality must then become asymptotically sharp. The associated scalar functions converge to affine functions in the same direction at both endpoints. Gaussian integration by parts gives the common Gaussian factorization. When $\kappa_0 = \kappa_1 = 1$, the full Poincaré inequality also gives a Γ_2 estimate after the translation component is removed. The entropy inequality follows from the standard variation formulas and a one-dimensional ODE comparison.

1.3 Relations to other methods and further directions

Poisson equations, Stein's method, and isoperimetry. The construction also has a Stein interpretation. For a probability measure $\mathrm{d}\mu = e^{-V}\,\mathrm{d}x$, the weighted divergence $-\operatorname{div}_\mu$ is a vector-valued Stein operator. Given a centered test function h, one seeks a field q satisfying

$$-\operatorname{div}_\mu q = h$$

and estimates its size or regularity. Such estimates are called Stein-factor bounds. Courtade, Fathi, and Pananjady constructed Stein kernels by a variational argument under a Poincaré inequality [15]. Mijoule, Raič, Reinert, and Swan introduced weak Stein equations and obtained weak Stein-factor bounds from a Poincaré inequality [42]. Fathi gave a different construction of Stein kernels through moment maps [24]. In this terminology, $\sqrt{C_P(\mu)}$ is the optimal L^2 norm of a right inverse of $-\operatorname{div}_\mu$ on centered functions. Our field q_t is an L^2 Stein solution for μ_t. In the usual application, Stein's method compares a measure with a reference law. Here one Poisson solution gives compatible fields at two prescribed endpoints. Their Bochner estimates control the L^2 norm of the interpolated field.

Jiang and Koskela used Poisson equations to derive local isoperimetric and Sobolev inequalities on metric measure spaces from estimates for the Cheeger–Poisson equation [29]. In particular, an estimate of the form

$$\|\nabla u\|_{L^\infty_{\mathrm{loc}}} \lesssim \|g\|_{L^{Q,1}_{\mathrm{loc}}} , \qquad \Delta u = g,$$

combined with integration by parts and the dual characterization of perimeter yields the isoperimetric inequality with the optimal exponent. Jiang, Koskela, and Yang later obtained local isoperimetric inequalities from Lipschitz regularity of Cheeger-harmonic functions [30]. Their argument uses an L^∞ gradient estimate to control perimeter. Our argument instead uses sharp L^2 estimates for two vector fields, one at each endpoint, to control variance at the intermediate measure.

Hessian metrics induced by optimal transport and related flows. Kolesnikov associates with the Brenier map $T = \nabla\Phi$ the Hessian metric $g = \operatorname{Hess}\Phi$. Its weighted Laplacian is

$$L_\Phi = \operatorname{div}_{\mu_0}\big((\operatorname{Hess}\Phi)^{-1}\nabla\big),$$

and the source–target integration-by-parts identity used below is stated in [36, Lemma 2.1 and (2.5)]. Klartag used a dual Bochner formula in his moment-map proof of Poincaré inequalities [32]. Kolesnikov and Milman derived Poincaré and logarithmic Sobolev inequalities by choosing Riemannian metrics adapted to the measure, including Hessian metrics arising from optimal transport [38].

Klartag and Putterman proved monotonicity of the Poincaré constant under Gaussian convolution, using a time-dependent diffusion in one proof and a contracting transport in another [34]. Serres studied the Poincaré constant along the Polchinski flow by dynamic Γ-calculus [47]. Those papers concern paths generated by the respective flows. Here the path is a prescribed displacement interpolation, while the Bochner estimates are taken at the two endpoints and linked by the differential of the Brenier map.

Our argument does not use a curvature bound for the Hessian metric. For $X = H^{-1}\nabla u$ and $Y \circ T = \nabla u$, the two matrix identities in Step 2 allow the Euclidean vector Bochner identity to be applied separately at the source and the target. They give

$$\kappa_0 \|X\|^2_{L^2(\mu_0)} \le \|g\|^2_{L^2(\mu_0)}, \qquad \kappa_1 \|Y\|^2_{L^2(\mu_1)} \le \|g\|^2_{L^2(\mu_0)}.$$

Combining the two separate estimates only at the final interpolation step gives the optimal coefficient in Theorem 1.1. It also allows us to analyze equality at both endpoints.

The matrix fact used to control $\mathrm{tr}((DX)^2)$ is the following: if H is positive definite and HDX is symmetric, then DX is similar to a symmetric matrix and $\mathrm{tr}((DX)^2) \ge 0$. The same conjugation appears in [33, proof of Lemma 3.1]; a related trace inequality for differentials of common-source Brenier interpolants appears in [2, proof of Theorem 1.5]. Cheng and Zhou proved that equality in the Bakry–Émery spectral-gap bound on a complete weighted manifold forces a Gaussian factor [10, Theorem 2]. Quantitative Euclidean stability was developed by De Philippis and Figalli [16, Section 4] and by Courtade and Fathi [14]. Here equality at one interior time forces both endpoint deficits to vanish and selects the same Gaussian direction at the two endpoints.

The entropy application builds on the second-variation argument of Aishwarya and Rotem [3, Lemma 4.3 and Theorem 4.6], formulated for even measures and even velocity potentials. Aishwarya and Li [2, Theorems 1.5 and 1.7] obtained an entropy-power inequality and a Gaussian Borell–Brascamp–Lieb inequality using a coupling constructed from a common Gaussian source. Unlike that construction, our entropy inequality is evaluated along the Wasserstein geodesic between the prescribed endpoints. Bolley, Gentil, and Guillin derived dimensional refinements of logarithmic Sobolev, Talagrand, and Brascamp–Lieb inequalities [4]. Eskenazis and Shenfeld developed intrinsic dimensional functional inequalities on model spaces [23].

Cordero-Erausquin and Eskenazis proved a functional dimensional Brunn–Minkowski inequality for rotationally invariant log-concave measures [11]. Malliaris, Melbourne, Roberto, and Roysdon obtained Gaussian Borell–Brascamp–Lieb inequalities through functional liftings of restricted geometric inequalities [40]. Our extension instead assumes that the probability measures proportional to $f\,\mathrm{d}\gamma$ and $g\,\mathrm{d}\gamma$ have the same barycenter. It is proved along the prescribed optimal transport interpolation. Kolesnikov and Milman derived a weighted Brunn–Minkowski inequality from a boundary Poincaré inequality along a geometric evolution [39]. Eskenazis, Giannopoulos, and Tziotziou proved a dimensional Brunn–Minkowski inequality for log-concave measures by a diffusion argument [21].

Wasserstein barycenters. We next ask whether Theorem 1.1 extends from two endpoints to Wasserstein barycenters. Let $\lambda_1, \dots, \lambda_N > 0$ with $\sum_{i=1}^N \lambda_i = 1$, and let $\bar{\mu}$ be a

minimizer of

$$\nu \longmapsto \sum_{i=1}^{N} \lambda_i W_2^2(\nu, \mu_i).$$

For $N = 2$, this reduces to a point on the Wasserstein geodesic, and Theorem 1.1 suggests the following question; see [1] for existence and basic properties of quadratic Wasserstein barycenters.

Question 1.7 (Wasserstein barycenter version)**.** Suppose that μ_i is κ_i-strongly log-concave for $i = 1, \dots, N$. Does their quadratic Wasserstein barycenter satisfy

$$\sqrt{C_P(\bar{\mu})} \le \sum_{i=1}^{N} \frac{\lambda_i}{\sqrt{\kappa_i}} \, ?$$

Can one also characterize equality by Gaussian factors in a common direction, with variances κ_i^{-1} for the respective input measures?

For two endpoints, the Hessian of one Brenier potential and one scalar Poisson equation produce a Bochner couple. With more than two endpoints, one would need compatible vector fields for several optimal transport maps and their Hessians. It is not clear how one Poisson equation could provide all the required estimates. We leave Question 1.7 open. Han and Liu proved that the Wasserstein Jensen inequality for Boltzmann entropy at barycenters of arbitrary finite families characterizes finite-dimensional normed spaces whose norm is induced by an inner product [27].

Riemannian version. In a companion work in preparation, entitled *Bochner Couples for Optimal Transport on Riemannian Manifolds*, the authors study the Riemannian analogue. There Jacobi fields describe the differential of the transport maps in place of the Euclidean identity $DT = \operatorname{Hess} \Phi$. A Codazzi-type tensor measures the failure of the matrix symmetry used here. No result from that preprint is used in the present paper.

Organization of the paper. Section 2 constructs a Bochner couple from a weighted Poisson equation and proves the interpolation estimate under smooth uniform regularity assumptions. Section 3 develops a curvature-preserving regularization and removes those auxiliary assumptions by stability of optimal transport. Section 4 proves the equality characterization and identifies the maximal common Gaussian factor of the endpoints. Section 5 applies the main theorem to centered Gaussian relative entropy, Gaussian Brunn–Minkowski inequalities with barycenter terms, and Gaussian functional inequalities after centering.

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2 Bochner couples from a weighted Poisson equation

We first prove the estimate under auxiliary uniform regularity assumptions on the optimal transport map. A weighted Poisson equation involving the Hessian of the Brenier potential produces a Bochner couple. Proposition 2.5 interpolates its two fields. Smoothness of

the endpoint potentials alone does not imply that the Brenier map is a smooth global diffeomorphism. The regularized potentials in Section 3 satisfy two-sided uniform Hessian bounds. Caffarelli's contraction theory then gives global bi-Lipschitz control. The regularity theorem of Cordero-Erausquin and Figalli for unbounded supports gives smoothness. The statement used in Lemma 3.3 follows directly from [35, Theorem 2.2] and [12, Remark 4 and Corollary 5].

Throughout this section, assume that V_0, V_1 are smooth and that $\mathrm{d}\mu_i = e^{-V_i}\,\mathrm{d}x$ with $\operatorname{Hess} V_i \succeq \kappa_i I$. Assume also that the Brenier map $T = \nabla\Phi$ from μ_0 to μ_1 is a smooth diffeomorphism. Finally, assume that $H = \operatorname{Hess}\Phi$ and H^{-1} are bounded in global operator norm. Equivalently, $\lambda I \preceq H \preceq \Lambda I$ for some $0 < \lambda \le \Lambda < \infty$. These are temporary hypotheses for the smooth argument and will be obtained from the regularization in Section 3. For symmetric matrices, $A \preceq B$ means that $B - A$ is positive semidefinite, and $\|A\|_{\mathrm{HS}}$ denotes the Hilbert–Schmidt norm.

For $\mathrm{d}\mu = e^{-V}\,\mathrm{d}x$, the weighted divergence with respect to μ is defined through integration by parts and is given by

$$\operatorname{div}_\mu X = \operatorname{div} X - \langle \nabla V, X\rangle\,. \tag{2.1}$$

We use the convention $(DX)_{ij} = \partial_j X_i$.

Lemma 2.1 (Weighted vector Bochner identity)**.** *For every $X \in C_c^\infty(\mathbb{R}^n;\mathbb{R}^n)$,*

$$\int (\operatorname{div}_\mu X)^2\,\mathrm{d}\mu = \int \left\{\operatorname{tr}((DX)^2) + \operatorname{Hess} V[X,X]\right\}\,\mathrm{d}\mu. \tag{2.2}$$

Proof. The weighted integration by parts yields

$$\int (\operatorname{div}_\mu X)^2\,\mathrm{d}\mu = -\int X_j \partial_j(\operatorname{div}_\mu X)\,\mathrm{d}\mu, \tag{2.3}$$

where repeated indices are summed. Expanding the derivative then implies

$$\int (\operatorname{div}_\mu X)^2\,\mathrm{d}\mu = -\int X_j\partial_i\partial_j X_i\,\mathrm{d}\mu + \int V_{ij}X_iX_j\,\mathrm{d}\mu + \int V_i X_j \partial_j X_i\,\mathrm{d}\mu. \tag{2.4}$$

Integrating the first term by parts in the i-th coordinate implies

$$-\int X_j\partial_i(\partial_j X_i)\,\mathrm{d}\mu = \int (\partial_i X_j - V_i X_j)\partial_j X_i\,\mathrm{d}\mu. \tag{2.5}$$

Combining (2.4) and (2.5) then yields (2.2). □

Remark 2.2. The first term in (2.2) is not, in general, the Hilbert–Schmidt norm of DX:

$$\operatorname{tr}((DX)^2) = \sum_{i,j} \partial_j X_i\,\partial_i X_j, \qquad \|DX\|_{\mathrm{HS}}^2 = \sum_{i,j} (\partial_j X_i)^2.$$

The two expressions agree when DX is symmetric, but $\operatorname{tr}((DX)^2)$ can have either sign for a general vector field. Thus (2.2) cannot be applied directly at an intermediate time: the potential of μ_t need not have a useful Hessian lower bound, and the trace term has no sign for a general vector field with the prescribed divergence. Proposition 2.4 instead works at the endpoints, where the curvature bounds are available, and uses the symmetry of HDX, with $H = \operatorname{Hess}\Phi$, to recover the nonnegativity of the trace terms.

Definition 2.3 (Smooth Bochner couple)**.** Let $T = \nabla\Phi$ transport μ_0 to μ_1 and set $H = \operatorname{Hess}\Phi$. Given a smooth mean-zero datum g and constants $a_0, a_1 \geq 0$, a pair of smooth vector fields (X, Y) is called a *Bochner couple* with datum g and constants (a_0, a_1) if the relations

$$Y \circ T = HX, \qquad -\operatorname{div}_{\mu_0} X = g, \qquad -(\operatorname{div}_{\mu_1} Y) \circ T = g, \tag{2.6}$$

hold, the two matrix symmetry conditions

$$HDX \text{ is symmetric}, \qquad ((DY) \circ T)H \text{ is symmetric} \tag{2.7}$$

hold, and the endpoint Bochner estimates

$$\|X\|_{L^2(\mu_0)} \leq a_0 \|g\|_{L^2(\mu_0)}, \qquad \|Y\|_{L^2(\mu_1)} \leq a_1 \|g\|_{L^2(\mu_0)} \tag{2.8}$$

are valid. If there is a scalar function u such that

$$X = H^{-1}\nabla u, \qquad Y \circ T = \nabla u \tag{2.9}$$

then we say that the couple is generated by the weighted Poisson equation. Here $H = DT = \operatorname{Hess}\Phi$ is the differential of the Brenier map that relates the two fields.

We call constructing a Bochner couple from common data and then interpolating its two fields *Bochner coupling*.

Proposition 2.4 (Endpoint estimates for the weighted Poisson equation)**.** *For every $g \in L^2(\mu_0)$ with $\int g \, \mathrm{d}\mu_0 = 0$, there exists a unique $u \in H^1(\mu_0)$ satisfying $\int u \, \mathrm{d}\mu_0 = 0$ and*

$$-\operatorname{div}_{\mu_0}(H^{-1}\nabla u) = g \tag{2.10}$$

in the weak sense. Setting $X = H^{-1}\nabla u$ and defining Y by $Y \circ T = \nabla u$, one also has

$$-(\operatorname{div}_{\mu_1} Y) \circ T = g \tag{2.11}$$

in the weak sense, together with the estimates

$$\|X\|_{L^2(\mu_0)} \leq \kappa_0^{-1/2} \|g\|_{L^2(\mu_0)}, \qquad \|Y\|_{L^2(\mu_1)} = \|\nabla u\|_{L^2(\mu_0)} \leq \kappa_1^{-1/2} \|g\|_{L^2(\mu_0)}. \tag{2.12}$$

Proof. **Step 1: Endpoint identities and Bochner estimates.** We begin with $u \in C_c^\infty(\mathbb{R}^n)$. Set

$$X = H^{-1}\nabla u, \qquad Bu = -\operatorname{div}_{\mu_0} X, \qquad Y(Tx) = \nabla u(x) = H(x)X(x).$$

Both fields are compactly supported, and we will verify that (X, Y) is a Bochner couple with datum Bu and constants $(\kappa_0^{-1/2}, \kappa_1^{-1/2})$. We first derive the endpoint estimates (the a priori estimates) with $g = Bu$; the extension to an arbitrary mean-zero $g \in L^2(\mu_0)$ is carried out at the end of the proof by a density argument. For every $\psi \in C_c^\infty(\mathbb{R}^n)$,

$$\int \langle \nabla\psi, Y \rangle \, \mathrm{d}\mu_1 = \int \langle \nabla\psi(Tx), H(x)X(x) \rangle \, \mathrm{d}\mu_0(x) = \int \langle \nabla(\psi \circ T), X \rangle \, \mathrm{d}\mu_0, \tag{2.13}$$

which implies

$$(\operatorname{div}_{\mu_1} Y) \circ T = \operatorname{div}_{\mu_0} X = -Bu \tag{2.14}$$

in the weak sense. This transformation rule for weighted divergence under the Brenier map appears in [36, Lemma 2.1 and (2.5)]. Let $P = DX$, so that $P_{kj} = \partial_j X_k$. Since $H_{ik} = (\operatorname{Hess}\Phi)_{ik} = \partial_i\partial_k\Phi$, differentiating $(HX)_i = \partial_i u$ with respect to x_j yields

$$(\operatorname{Hess} u)_{ij} = (\partial_j H_{ik})X_k + H_{ik}P_{kj} = \Phi_{ikj}X_k + H_{ik}P_{kj}, \tag{2.15}$$

and hence

$$\operatorname{Hess} u = C_X + HP, \qquad (C_X)_{ij} = \Phi_{ijk}X_k. \tag{2.16}$$

The smoothness of Φ implies that C_X is symmetric. Since $\operatorname{Hess} u$ is symmetric as well, (2.16) shows that $HP = \operatorname{Hess} u - C_X$ is symmetric. The matrix

$$H^{-1/2}(HP)H^{-1/2} = H^{1/2}PH^{-1/2}$$

is symmetric and similar to P. Hence,

$$\operatorname{tr}(P^2) = \left\|H^{-1/2}(HP)H^{-1/2}\right\|_{\mathrm{HS}}^2 \geq 0. \tag{2.17}$$

Applying Lemma 2.1 with μ_0 and using (2.17), we obtain

$$\|X\|_{L^2(\mu_0)} = \left\|H^{-1}\nabla u\right\|_{L^2(\mu_0)} \leq \kappa_0^{-1/2}\left\|Bu\right\|_{L^2(\mu_0)} = \kappa_0^{-1/2}\left\|g\right\|_{L^2(\mu_0)}. \tag{2.18}$$

At the target, let $Q(x) = (D_y Y)(Tx)$. Since $Y(Tx) = \nabla u(x)$, we have $QH = \operatorname{Hess} u$. Thus Q is similar to the symmetric matrix $H^{-1/2}\operatorname{Hess} u H^{-1/2}$, and hence

$$\operatorname{tr}(Q^2) = \left\|H^{-1/2}\operatorname{Hess} u H^{-1/2}\right\|_{\mathrm{HS}}^2 \geq 0. \tag{2.19}$$

Applying Lemma 2.1 with μ_1, together with (2.14) and (2.19), we obtain

$$\|Y\|_{L^2(\mu_1)} = \|\nabla u\|_{L^2(\mu_0)} \leq \kappa_1^{-1/2}\left\|Bu\right\|_{L^2(\mu_0)}. \tag{2.20}$$

Thus the two bounds are the endpoint estimates of a Bochner couple with constants $(\kappa_0^{-1/2}, \kappa_1^{-1/2})$.

Step 2: Weak solvability and range density. The existence and uniqueness of a centered weak solution of (2.10) already follow from the Lax–Milgram theorem [7, Corollary 5.8], applied on the mean-zero subspace of $H^1(\mu_0)$ to the bilinear form

$$(v, w) \longmapsto \int \left\langle H^{-1}\nabla v, \nabla w\right\rangle \mathrm{d}\mu_0.$$

Here boundedness and coercivity follow from the global bounds on H and H^{-1} and the Poincaré inequality for μ_0. To ensure that this solution also inherits (2.18) and (2.20), it remains to prove that the range of the operator below is dense.

Let $B = -\operatorname{div}_{\mu_0}(H^{-1}\nabla)$ on $C_c^\infty(\mathbb{R}^n)$. Its range is dense in

$$L_0^2(\mu_0) = \left\{h \in L^2(\mu_0) : \int h\,\mathrm{d}\mu_0 = 0\right\}.$$

Indeed, if $h \in L_0^2(\mu_0)$ is orthogonal to the range and $a = e^{-V_0}H^{-1}$, then

$$-\int h\operatorname{div}(a\nabla\varphi)\,\mathrm{d}x = 0, \qquad \varphi \in C_c^\infty(\mathbb{R}^n). \tag{2.21}$$

The operator $-\operatorname{div}(a\nabla)$ is smooth and locally uniformly elliptic, so standard interior elliptic regularity [28] yields $h \in H^1_{\mathrm{loc}}$ (indeed, $h \in C^\infty$). Choosing a test function $\eta_R^2 h$, where $\eta_R = 1$ on B_R, $\operatorname{supp}\eta_R \subseteq B_{2R}$, and $|\nabla\eta_R| \le C/R$, gives

$$\begin{aligned}\int \eta_R^2 \left\langle H^{-1}\nabla h, \nabla h\right\rangle \mathrm{d}\mu_0 &= -2\int \eta_R h \left\langle H^{-1}\nabla h, \nabla\eta_R\right\rangle \mathrm{d}\mu_0 \\ &\le \frac{1}{2}\int \eta_R^2 \left\langle H^{-1}\nabla h, \nabla h\right\rangle \mathrm{d}\mu_0 + 2\int h^2 \left\langle H^{-1}\nabla\eta_R, \nabla\eta_R\right\rangle \mathrm{d}\mu_0.\end{aligned} \tag{2.22}$$

Combining with $H^{-1} \preceq \lambda^{-1} I$, we obtain

$$\int \eta_R^2 \left\langle H^{-1}\nabla h, \nabla h\right\rangle \mathrm{d}\mu_0 \le \frac{4C^2}{\lambda R^2}\|h\|^2_{L^2(\mu_0)}. \tag{2.23}$$

Letting $R \to \infty$ on each fixed ball yields $\nabla h = 0$; the zero-mean condition then implies $h = 0$. Thus the range is dense.

Step 3: Approximation and passage to the limit. Choose $u_k^0 \in C_c^\infty(\mathbb{R}^n)$ such that $Bu_k^0 \to g$ in $L^2(\mu_0)$, and set $u_k = u_k^0 - \int u_k^0 \mathrm{d}\mu_0$. Then u_k is centered, while $\nabla u_k = \nabla u_k^0$ and $Bu_k = Bu_k^0$. Applying (2.18) and (2.20) to $u_k^0 - u_\ell^0 \in C_c^\infty(\mathbb{R}^n)$ shows that ∇u_k and $H^{-1}\nabla u_k$ are Cauchy in $L^2(\mu_0)$. Moreover, since each u_k is centered, the endpoint Poincaré inequality and the Cauchy property of (∇u_k) show that (u_k) is Cauchy in $H^1(\mu_0)$. It follows that $u_k \to u$ in $H^1(\mu_0)$ for some centered $u \in H^1(\mu_0)$. For every $\varphi \in C_c^\infty(\mathbb{R}^n)$,

$$\int \left\langle H^{-1}\nabla u, \nabla\varphi\right\rangle \mathrm{d}\mu_0 = \lim_{k\to\infty}\int \left\langle H^{-1}\nabla u_k, \nabla\varphi\right\rangle \mathrm{d}\mu_0 = \lim_{k\to\infty}\int (Bu_k)\varphi \,\mathrm{d}\mu_0 = \int g\varphi \,\mathrm{d}\mu_0. \tag{2.24}$$

Thus u is a centered weak solution of (2.10). By the density of $C_c^\infty(\mathbb{R}^n)$ in $H^1(\mu_0)$, the weak identity extends to all $H^1(\mu_0)$ test functions. Hence u coincides with the unique solution provided by the Lax–Milgram theorem. Define Y_k by $Y_k \circ T = \nabla u_k$. Since $u_k \to u$ in $H^1(\mu_0)$, we have $Y_k \to Y$ in $L^2(\mu_1)$. Applying (2.13) to u_k^0 and then passing to the limit, we obtain

$$\int \langle \nabla\psi, Y\rangle \mathrm{d}\mu_1 = \int g\,\psi\circ T \,\mathrm{d}\mu_0, \qquad \psi \in C_c^\infty(\mathbb{R}^n), \tag{2.25}$$

which implies $-(\operatorname{div}_{\mu_1} Y)\circ T = g$ in the weak sense. Passing to the limit in (2.18) and (2.20) yields (2.12). □

At the smooth level, the same Poisson solution u gives both matrix identities. The matrix HDX is symmetric at the source, and $((D_y Y)\circ T)H = \operatorname{Hess} u$ is symmetric at the target. These facts give the sharp endpoint estimates. They do not require a curvature bound for any intermediate measure. Together with (2.14), these estimates provide the input for the affine interpolation in Proposition 2.5.

Proposition 2.5 (Poincaré estimate in the smooth case)**.** *Assume that $V_i \in C^\infty(\mathbb{R}^n)$ and $\operatorname{Hess} V_i \succeq \kappa_i I$ for $i = 0, 1$. Suppose that the Brenier map $T = \nabla\Phi$ from μ_0 to μ_1 is a smooth diffeomorphism and that $\lambda I \preceq \operatorname{Hess}\Phi \preceq \Lambda I$ for some $0 < \lambda \le \Lambda < \infty$. Then (1.1) holds for every $t \in [0, 1]$.*

Proof. Fix $f \in C_c^\infty(\mathbb{R}^n)$ and let $h = f - \int f \,\mathrm{d}\mu_t$. Set

$$A_t = DF_t = (1-t)I + tH, \qquad \alpha_t = (1-t) + t\lambda, \qquad g = h\circ F_t.$$

Since $(F_t)_{\#}\mu_0 = \mu_t$, we have $g \in L^2_0(\mu_0)$ and $\|g\|_{L^2(\mu_0)} = \|h\|_{L^2(\mu_t)}$. The uniform lower bound $A_t \succeq \alpha_t I$ yields

$$\big(F_t(x) - F_t(y)\big) \cdot (x - y) \geq \alpha_t |x - y|^2, \tag{2.26}$$

which implies that F_t is injective and proper. The local inverse function theorem makes its image open, while properness makes the image closed; hence F_t is a global diffeomorphism.

Apply Proposition 2.4 to $g = h \circ F_t$ and denote $X = H^{-1}\nabla u$. Define q at time t by

$$q(F_t(x)) = A_t(x)X(x) = (1-t)X(x) + t\nabla u(x).$$

For $\psi \in C_c^\infty(\mathbb{R}^n)$, properness of F_t ensures that $\psi \circ F_t$ is compactly supported. Using $(F_t)_{\#}\mu_0 = \mu_t$, the definition $q \circ F_t = DF_t\, X$, and the chain rule, we obtain

$$\begin{aligned} \int q \cdot \nabla\psi \, \mathrm{d}\mu_t &= \int q(F_t(x)) \cdot \nabla\psi(F_t(x)) \, \mathrm{d}\mu_0(x) \\ &= \int X \cdot \nabla(\psi \circ F_t) \, \mathrm{d}\mu_0 \\ &\overset{*}{=} \int g\, \psi \circ F_t \, \mathrm{d}\mu_0 = \int h\psi \, \mathrm{d}\mu_t, \end{aligned} \tag{2.27}$$

where $(*)$ follows from the weak formulation of $-\operatorname{div}_{\mu_0} X = g$. Hence $-\operatorname{div}_{\mu_t} q = h$ in the weak sense. Moreover, Proposition 2.4 gives

$$\|q\|_{L^2(\mu_t)} \leq (1-t)\,\|X\|_{L^2(\mu_0)} + t\,\|\nabla u\|_{L^2(\mu_0)} \leq \left(\frac{1-t}{\sqrt{\kappa_0}} + \frac{t}{\sqrt{\kappa_1}}\right) \|h\|_{L^2(\mu_t)}. \tag{2.28}$$

Since $h = f - \int f \, \mathrm{d}\mu_t$, taking $\psi = f$ in (2.27), we obtain

$$\operatorname{Var}_{\mu_t}(f) = \int q \cdot \nabla f \, \mathrm{d}\mu_t \leq \left(\frac{1-t}{\sqrt{\kappa_0}} + \frac{t}{\sqrt{\kappa_1}}\right) \sqrt{\operatorname{Var}_{\mu_t}(f)} \left(\int |\nabla f|^2 \, \mathrm{d}\mu_t\right)^{1/2}, \tag{2.29}$$

which proves the proposition. □

3 Regularization and transport stability

To pass to nonsmooth endpoint potentials, we need a smooth approximation that preserves their curvature lower bounds. Gaussian convolution alone weakens these bounds, so we combine it with a compensating dilation. The covariance estimate needed to verify this regularization is the following linear form of the Brascamp–Lieb inequality.

Lemma 3.1. *Let $U : \mathbb{R}^n \to (-\infty, +\infty]$ be proper and lower semicontinuous, with $U - \lambda|x|^2/2$ convex for some $\lambda > 0$. If $0 < \int e^{-U} \, \mathrm{d}x < \infty$, $\mathrm{d}\nu \propto e^{-U} \, \mathrm{d}x$, and ζ has law ν, then*

$$\operatorname{Cov}(\zeta) \preceq \lambda^{-1} I. \tag{3.1}$$

Proof. Write $U = \lambda|x|^2/2 + W$, where W is proper, lower semicontinuous, and convex. Choose an affine minorant $W(x) \geq p \cdot x + b$, and let

$$W_m(x) = \inf_{y \in \mathbb{R}^n} \left\{ W(y) + \frac{m}{2}|x - y|^2 \right\}, \qquad m \geq 1,$$

be the Moreau envelopes of W. They are finite and convex, $W_m \uparrow W$ pointwise, and

$$W_m(x) \geq p \cdot x + b - \frac{|p|^2}{2m}. \tag{3.2}$$

Let ρ be a smooth, compactly supported, radial probability density, write $\rho_\delta(x) = \delta^{-n}\rho(x/\delta)$, and set $W_{m,\delta} = W_m * \rho_\delta$. Then $W_{m,\delta}$ is smooth and convex. Because $\int z\rho_\delta(z)\,\mathrm{d}z = 0$, (3.2) also holds with $W_{m,\delta}$ in place of W_m.

The classical Brascamp–Lieb inequality [6], applied to the smooth potential $\lambda|x|^2/2 + W_{m,\delta}$, gives

$$\mathrm{Var}_{\nu_{m,\delta}}(a \cdot x) \leq \lambda^{-1}|a|^2, \qquad a \in \mathbb{R}^n, \tag{3.3}$$

where $\mathrm{d}\nu_{m,\delta} \propto e^{-\lambda|x|^2/2 - W_{m,\delta}(x)}\,\mathrm{d}x$. For fixed m, the finite convex function W_m is continuous and $W_{m,\delta} \to W_m$ locally uniformly as $\delta \downarrow 0$. Moreover, (3.2), uniformly for $m \geq 1$ and $\delta > 0$, bounds all the corresponding unnormalized densities by a constant times $e^{-\lambda|x|^2/2 - p\cdot x}$. This function remains integrable after multiplication by $1 + |x|^2$, so dominated convergence applies to the normalization constants and to the first and second moments. We may first let $\delta \downarrow 0$ in (3.3) and then let $m \to \infty$. Since $W_m \uparrow W$, this yields

$$\mathrm{Var}_{\nu}(a \cdot x) \leq \lambda^{-1}|a|^2, \qquad a \in \mathbb{R}^n,$$

which is equivalent to the stated covariance bound. □

Lemma 3.2. *Let $\kappa > 0$, and let μ have potential V with $V - \kappa|x|^2/2$ convex. Let $\xi \sim \mu$ and $G \sim N(0, I)$ be independent. For $\varepsilon > 0$, define*

$$\xi^\varepsilon = \frac{\xi + \sqrt{\varepsilon}\,G}{\sqrt{1+\kappa\varepsilon}}, \qquad \mu^\varepsilon = \mathrm{Law}(\xi^\varepsilon).$$

Then $\mu^\varepsilon \to \mu$ in W_2. The measure μ^ε has a smooth positive density e^{-V^ε} whose potential satisfies

$$\kappa I \preceq \mathrm{Hess}\, V^\varepsilon \preceq \frac{1+\kappa\varepsilon}{\varepsilon} I. \tag{3.4}$$

The lower Hessian bound is also the classical convolution rule for strong log-concavity; see [46, Proposition 5.5]. We include the conditional-moment proof because it simultaneously gives the upper Hessian bound and covers extended-valued endpoint potentials.

Proof. **Step 1: Convergence in W_2.** Set $a_\varepsilon = \sqrt{1+\kappa\varepsilon}$. Writing $V = \kappa|x|^2/2 + W$ with W convex, choose an affine minorant $W(x) \geq p \cdot x + b$. Then $e^{-V(x)}$ is bounded by a constant times $\exp(-\kappa|x|^2/2 - p\cdot x)$, and hence $\mathbb{E}|\xi|^2 < \infty$. Couple μ^ε and μ by using the same pair (ξ, G). By the definition of W_2,

$$\begin{aligned} W_2^2(\mu^\varepsilon, \mu) &\leq \mathbb{E}\left|(a_\varepsilon^{-1} - 1)\xi + \frac{\sqrt{\varepsilon}}{a_\varepsilon} G\right|^2 \\ &= (a_\varepsilon^{-1} - 1)^2 \mathbb{E}|\xi|^2 + \frac{\varepsilon}{a_\varepsilon^2}\mathbb{E}|G|^2 = (a_\varepsilon^{-1} - 1)^2 \mathbb{E}|\xi|^2 + \frac{\varepsilon n}{a_\varepsilon^2} \longrightarrow 0, \end{aligned} \tag{3.5}$$

where the cross term vanishes since G is independent of ξ and $\mathbb{E}[G] = 0$. This proves $\mu^\varepsilon \to \mu$ in W_2.

Step 2: Hessian bounds after Gaussian smoothing. Set $Y_\varepsilon = \xi + \sqrt{\varepsilon}G$, let ϕ_ε be the density of $N(0, \varepsilon I)$, and write

$$p_\varepsilon(y) = \int \phi_\varepsilon(y - x)\,\mathrm{d}\mu(x) = e^{-U_\varepsilon(y)}. \tag{3.6}$$

This density is smooth and strictly positive. For each $y \in \mathbb{R}^n$, Bayes' rule gives

$$\mathbb{P}(\xi \in \mathrm{d}x \mid Y_\varepsilon = y) = \frac{\phi_\varepsilon(y-x)}{p_\varepsilon(y)}\,\mathrm{d}\mu(x). \tag{3.7}$$

Define $m_\varepsilon(y) = \mathbb{E}[\xi \mid Y_\varepsilon = y]$. Then, for $j = 1, \dots, n$,

$$p_\varepsilon(y) m_{\varepsilon,j}(y) = \int x_j \phi_\varepsilon(y-x)\,\mathrm{d}\mu(x). \tag{3.8}$$

For $1 \le j, k \le n$, differentiation of (3.6) and (3.8) yields

$$\partial_k p_\varepsilon(y) = -\varepsilon^{-1} \int (y_k - x_k)\phi_\varepsilon(y-x)\,\mathrm{d}\mu(x), \tag{3.9}$$

and

$$\partial_k (p_\varepsilon(y) m_{\varepsilon,j}(y)) = -\varepsilon^{-1} \int x_j (y_k - x_k)\phi_\varepsilon(y-x)\,\mathrm{d}\mu(x). \tag{3.10}$$

The quotient rule therefore yields

$$\nabla \log p_\varepsilon(y) = -\varepsilon^{-1}\big(y - m_\varepsilon(y)\big), \qquad Dm_\varepsilon(y) = \varepsilon^{-1}\operatorname{Cov}(\xi \mid Y_\varepsilon = y). \tag{3.11}$$

Indeed, the (j,k) entry in the second identity is $\varepsilon^{-1}\left(\mathbb{E}[\xi_j \xi_k \mid Y_\varepsilon = y] - m_{\varepsilon,j}(y) m_{\varepsilon,k}(y)\right)$. Since $\mathbb{E}|\xi|^2 < \infty$, differentiation under the integral sign is justified by dominated convergence, locally uniformly in y. The first identity in (3.11) is the Gaussian Tweedie formula. The second expresses the derivative of the conditional mean in terms of the conditional covariance. See Efron [19] for the statistical formulation and Dytso, Poor, and Shamai [18] for vector Gaussian derivative identities. Since $U_\varepsilon = -\log p_\varepsilon$, differentiating the first identity in (3.11) and substituting the formula for Dm_ε yields

$$\operatorname{Hess} U_\varepsilon(y) = \varepsilon^{-1} I - \varepsilon^{-2}\operatorname{Cov}(\xi \mid Y_\varepsilon = y). \tag{3.12}$$

Thus, up to an additive normalizing constant, the conditional law above has potential

$$x \longmapsto V(x) + \frac{|y-x|^2}{2\varepsilon},$$

which is $(\kappa + \varepsilon^{-1})$-strongly convex in the extended-valued sense. Therefore, by Lemma 3.1,

$$\operatorname{Cov}(\xi \mid Y_\varepsilon = y) \preceq \frac{\varepsilon}{1+\kappa\varepsilon} I. \tag{3.13}$$

Substituting (3.13) into (3.12) gives the lower Hessian bound, while $\operatorname{Cov}(\xi \mid Y_\varepsilon = y) \succeq 0$ gives the upper bound:

$$\frac{\kappa}{1+\kappa\varepsilon} I \preceq \operatorname{Hess} U_\varepsilon \preceq \varepsilon^{-1} I. \tag{3.14}$$

Finally, since $\xi^\varepsilon = Y_\varepsilon / a_\varepsilon$ and $a_\varepsilon^2 = 1 + \kappa\varepsilon$, the change of variables yields

$$V^\varepsilon(x) = U_\varepsilon(a_\varepsilon x) - n \log a_\varepsilon, \qquad \operatorname{Hess} V^\varepsilon(x) = a_\varepsilon^2 \operatorname{Hess} U_\varepsilon(a_\varepsilon x), \tag{3.15}$$

which proves (3.4). □

Lemma 3.3 (Regularity and bounds for the Brenier map)**.** *Let $\mathrm{d}\mu_i = e^{-V_i}\,\mathrm{d}x$, $i = 0, 1$, be smooth positive probability densities whose potentials satisfy*

$$\lambda_i^- I \preceq \operatorname{Hess} V_i \preceq \lambda_i^+ I, \qquad 0 < \lambda_i^- \le \lambda_i^+ < \infty.$$

Then the Brenier map $T = \nabla\Phi$ from μ_0 to μ_1 is a smooth global diffeomorphism and

$$\sqrt{\frac{\lambda_0^-}{\lambda_1^+}}\, I \preceq \operatorname{Hess}\Phi \preceq \sqrt{\frac{\lambda_0^+}{\lambda_1^-}}\, I. \tag{3.16}$$

Proof. Since the endpoint densities are smooth and positive on $\mathbb{R}^n$, they and their reciprocals are bounded on every compact set. Hence [12, Corollary 5], applied for every order of regularity, shows that $T = \nabla\Phi$ is a smooth global diffeomorphism. Moreover, its inverse is the reverse Brenier map $S : \mu_1 \to \mu_0$; see [12, Remark 4].

Caffarelli's contraction theorem [9], in the two-potential form of [35, Theorem 2.2], applied first to T and then to S, yields

$$\mathrm{Lip}(T) \le \sqrt{\frac{\lambda_0^+}{\lambda_1^-}}, \qquad \mathrm{Lip}(S) \le \sqrt{\frac{\lambda_1^+}{\lambda_0^-}}. \tag{3.17}$$

Since $S = T^{-1}$ and both maps are smooth, we have $DS(T(x)) = DT(x)^{-1}$. The preceding Lipschitz bounds therefore imply

$$\sqrt{\frac{\lambda_0^-}{\lambda_1^+}}\, I \preceq DT = \mathrm{Hess}\,\Phi \preceq \sqrt{\frac{\lambda_0^+}{\lambda_1^-}}\, I, \tag{3.18}$$

which proves (3.16). □

Proof of Theorem 1.1. Apply Lemma 3.2 to both endpoints with their respective curvature constants. Let $\varepsilon_j \downarrow 0$, denote the regularized endpoints by μ_i^j, $i = 0, 1$, and let π_j be their quadratic optimal plans. By Lemma 3.2 and Lemma 3.3, the assumptions of Proposition 2.5 hold with the same lower curvature constants κ_i.

By the stability theorem for optimal transport plans [48, Theorem 5.20], every weak subsequential limit of (π_j) is optimal between μ_0 and μ_1. Brenier's theorem [8] gives a unique quadratic optimal plan π because $\mu_0 \ll \mathcal{L}^n$. Hence $\pi_j \to \pi$ weakly. The marginals converge in W_2, so the joint second moments also converge. It follows from [48, Theorem 6.9] that $\pi_j \to \pi$ in $W_2(\mathbb{R}^n \times \mathbb{R}^n)$. Fix $t \in [0, 1]$ and let $L_t(x, y) = (1 - t)x + ty$. Since L_t is Lipschitz,

$$\mu_t^j := (L_t)_\# \pi_j \to (L_t)_\# \pi = \mu_t \quad \text{in } W_2. \tag{3.19}$$

For $f \in C_c^\infty(\mathbb{R}^n)$, by Proposition 2.5, we obtain

$$\mathrm{Var}_{\mu_t^j}(f) \le \left(\frac{1 - t}{\sqrt{\kappa_0}} + \frac{t}{\sqrt{\kappa_1}} \right)^2 \int |\nabla f|^2 \, \mathrm{d}\mu_t^j. \tag{3.20}$$

Since f, f^2, and $|\nabla f|^2$ are bounded and continuous, passing to the limit proves (1.1). Since $t \in [0, 1]$ was arbitrary, the conclusion holds along the entire geodesic. □

4 Gaussian rigidity

Theorem 1.4 states that equality at an interior time is equivalent to a common Gaussian factor. The variances of the two Gaussian factors are κ_i^{-1}. We first prove a quantitative rigidity lemma at one endpoint. It shows that a near-extremizer of the Poincaré inequality has an almost constant gradient. We apply this lemma to regularized near-extremizers for μ_t. The resulting limits are affine in the same direction at both endpoints, which gives the common Gaussian factorization. The converse follows by splitting the optimal transport along this Gaussian direction. Quantitative affine approximation of Poincaré near-extremizers for strongly log-concave measures was developed by De Philippis and Figalli [16, Section 4]. Courtade and Fathi proved an $O(\varepsilon)$ squared L^2-gradient estimate for normalized near-extremizers [14, Lemma 2.2]. The next lemma is its rescaled, homogeneous one-function form; we include the short derivation to make the scaling explicit.

Lemma 4.1 (Homogeneous form of quantitative Poincaré stability)**.** *Let ν be the probability measure $\mathrm{d}\nu = e^{-W}\,\mathrm{d}x$, where W is smooth and*

$$\operatorname{Hess} W \succeq \kappa I, \qquad \kappa > 0.$$

If $h \in H^1(\nu)$ satisfies $\int h\,\mathrm{d}\nu = 0$, set

$$D = \int |\nabla h|^2\,\mathrm{d}\nu - \kappa \int h^2\,\mathrm{d}\nu, \qquad a = \int \nabla h\,\mathrm{d}\nu.$$

Then $D \geq 0$ and

$$\int |\nabla h - a|^2\,\mathrm{d}\nu \leq 9D. \tag{4.1}$$

Proof. Since $\int h\,\mathrm{d}\nu = 0$, the Brascamp–Lieb inequality implies $D \geq 0$. The assertion is trivial when $h = 0$, so we may assume $s := \|h\|_{L^2(\nu)} > 0$.

Define $\tilde{\nu} = (x \mapsto \sqrt{\kappa}x)_{\#}\nu$ and $u(z) = s^{-1}h(z/\sqrt{\kappa})$. Then $\tilde{\nu}$ is 1-strongly log-concave and

$$\int u\,\mathrm{d}\tilde{\nu} = 0, \qquad \int u^2\,\mathrm{d}\tilde{\nu} = 1, \qquad \int |\nabla u|^2\,\mathrm{d}\tilde{\nu} = 1 + \frac{D}{\kappa s^2}. \tag{4.2}$$

Applying [14, Lemma 2.2] with $k = 1$, we obtain a unit vector $w \in \mathbb{R}^n$ such that

$$\int |\nabla u - w|^2\,\mathrm{d}\tilde{\nu} \leq \frac{9D}{\kappa s^2}. \tag{4.3}$$

Since $a = \int \nabla h\,\mathrm{d}\nu$ minimizes $c \longmapsto \int |\nabla h - c|^2\,\mathrm{d}\nu$ over $c \in \mathbb{R}^n$,

$$\int |\nabla h - a|^2\,\mathrm{d}\nu \leq \int |\nabla h - \sqrt{\kappa}s\,w|^2\,\mathrm{d}\nu = \kappa s^2 \int |\nabla u - w|^2\,\mathrm{d}\tilde{\nu} \leq 9D. \tag{4.4}$$

□

Proof of Theorem 1.4. **Step 1: Alignment of the vector fields.** Fix $t \in (0,1)$ and first suppose

$$C_P(\mu_t) = \left(\frac{1-t}{\sqrt{\kappa_0}} + \frac{t}{\sqrt{\kappa_1}}\right)^2. \tag{4.5}$$

Choose $\varphi_j \in C_c^\infty(\mathbb{R}^n)$ such that

$$\operatorname{Var}_{\mu_t}(\varphi_j) = 1, \qquad \int |\nabla \varphi_j|^2\,\mathrm{d}\mu_t \to \left(\frac{1-t}{\sqrt{\kappa_0}} + \frac{t}{\sqrt{\kappa_1}}\right)^{-2}. \tag{4.6}$$

For $\varepsilon > 0$, let μ_i^ε be the regularization of μ_i given by Lemma 3.2, and let μ_t^ε be the corresponding interpolated measure. By the stability argument in the proof of Theorem 1.1, $\mu_t^\varepsilon \to \mu_t$ in W_2 as $\varepsilon \to 0$. Since φ_j, φ_j^2, and $|\nabla\varphi_j|^2$ are bounded and continuous, we may choose $0 < \varepsilon_j < 1/j$ such that

$$\left|\operatorname{Var}_{\mu_t^{\varepsilon_j}}(\varphi_j) - \operatorname{Var}_{\mu_t}(\varphi_j)\right| \leq \frac{1}{j}, \qquad \left|\int |\nabla\varphi_j|^2\,\mathrm{d}\mu_t^{\varepsilon_j} - \int |\nabla\varphi_j|^2\,\mathrm{d}\mu_t\right| \leq \frac{1}{j}. \tag{4.7}$$

Denote the regularized endpoints by $\mu_i^j := \mu_i^{\varepsilon_j}$, $i = 0, 1$, and let T_j be the Brenier map from μ_0^j to μ_1^j. Denote $C_t = (1-t)\kappa_0^{-1/2} + t\kappa_1^{-1/2}$ and set

$$H_j = DT_j, \qquad F_j = (1-t)I + tT_j, \qquad \mu_t^j := \mu_t^{\varepsilon_j} = (F_j)_{\#}\mu_0^j.$$

For all sufficiently large j, define $f_j := \big(\varphi_j - \int \varphi_j \,\mathrm{d}\mu_t^j\big)/\sqrt{\mathrm{Var}_{\mu_t^j}(\varphi_j)}$. Then we have

$$\int f_j \,\mathrm{d}\mu_t^j = 0, \qquad \int |f_j|^2 \,\mathrm{d}\mu_t^j = 1, \qquad \int |\nabla f_j|^2 \,\mathrm{d}\mu_t^j \to C_t^{-2}. \tag{4.8}$$

Set $g_j = f_j \circ F_j$ and $r_j = (\nabla f_j) \circ F_j$. Then we have

$$\int g_j \,\mathrm{d}\mu_0^j = 0, \qquad \int |g_j|^2 \,\mathrm{d}\mu_0^j = 1, \qquad \int |r_j|^2 \,\mathrm{d}\mu_0^j \to C_t^{-2}. \tag{4.9}$$

Applying Proposition 2.4 to g_j, we obtain a centered function $u_j \in H^1(\mu_0^j)$. Set $X_j = H_j^{-1}\nabla u_j$ and $\mathsf{Y}_j = \nabla u_j$. Thus Y_j is the target field from Proposition 2.4 pulled back by T_j, and

$$-\operatorname{div}_{\mu_0^j} X_j = g_j, \qquad \|X_j\|_{L^2(\mu_0^j)} \le \kappa_0^{-1/2}, \qquad \|\mathsf{Y}_j\|_{L^2(\mu_0^j)} \le \kappa_1^{-1/2}. \tag{4.10}$$

All norms and inner products below are taken in $L^2(\mu_0^j)$. Since F_j is bi-Lipschitz, $g_j \in H^1(\mu_0^j)$ and is hence admissible as a test function in (4.10). Since $\nabla g_j = ((1-t)I + tH_j)r_j$,

$$\begin{aligned} 1 = \int g_j^2 \,\mathrm{d}\mu_0^j &\overset{(4.10)}{=} \int X_j \cdot \nabla g_j \,\mathrm{d}\mu_0^j = \int r_j \cdot \big((1-t)X_j + t\mathsf{Y}_j\big) \,\mathrm{d}\mu_0^j \\ &\le \|r_j\|_{L^2(\mu_0^j)} \,\|(1-t)X_j + t\mathsf{Y}_j\|_{L^2(\mu_0^j)} \\ &\le \|r_j\|_{L^2(\mu_0^j)} \,\big((1-t)\,\|X_j\|_{L^2(\mu_0^j)} + t\,\|\mathsf{Y}_j\|_{L^2(\mu_0^j)}\big) \\ &\le C_t \,\|r_j\|_{L^2(\mu_0^j)} \overset{(4.8)}{\to} 1, \end{aligned} \tag{4.11}$$

The following chain is therefore asymptotically sharp, with $\|r_j\|_{L^2(\mu_0^j)}^{-1} \to C_t$:

$$\|r_j\|_{L^2(\mu_0^j)}^{-1} \le \|(1-t)X_j + t\mathsf{Y}_j\|_{L^2(\mu_0^j)} \le (1-t)\,\|X_j\|_{L^2(\mu_0^j)} + t\,\|\mathsf{Y}_j\|_{L^2(\mu_0^j)} \le C_t. \tag{4.12}$$

To make explicit which inequalities become sharp, set

$$\begin{aligned} \delta_j^{\mathrm{CS}} &:= \|(1-t)X_j + t\mathsf{Y}_j\|_{L^2(\mu_0^j)} - \|r_j\|_{L^2(\mu_0^j)}^{-1}, \\ \delta_j^{\triangle} &:= (1-t)\,\|X_j\|_{L^2(\mu_0^j)} + t\,\|\mathsf{Y}_j\|_{L^2(\mu_0^j)} - \|(1-t)X_j + t\mathsf{Y}_j\|_{L^2(\mu_0^j)}, \\ \delta_j^{\mathrm{end}} &:= C_t - (1-t)\,\|X_j\|_{L^2(\mu_0^j)} - t\,\|\mathsf{Y}_j\|_{L^2(\mu_0^j)}. \end{aligned}$$

These three quantities are nonnegative, and

$$\delta_j^{\mathrm{CS}} + \delta_j^{\triangle} + \delta_j^{\mathrm{end}} = C_t - \|r_j\|_{L^2(\mu_0^j)}^{-1} \longrightarrow 0.$$

In particular, $\delta_j^{\mathrm{end}} \to 0$. Since $0 < t < 1$ and the endpoint norms are bounded above by $\kappa_0^{-1/2}$ and $\kappa_1^{-1/2}$, respectively, we have

$$\delta_j^{\mathrm{end}} = (1-t)\big(\kappa_0^{-1/2} - \|X_j\|_{L^2(\mu_0^j)}\big) + t\big(\kappa_1^{-1/2} - \|\mathsf{Y}_j\|_{L^2(\mu_0^j)}\big),$$

and each term on the right is nonnegative. Hence

$$\|X_j\|_{L^2(\mu_0^j)} \to \kappa_0^{-1/2}, \qquad \|\mathsf{Y}_j\|_{L^2(\mu_0^j)} \to \kappa_1^{-1/2}. \tag{4.13}$$

Expanding $\|(1-t)X_j + t\mathsf{Y}_j\|_{L^2(\mu_0^j)}^2$ and combining with (4.13) yields

$$\langle X_j, \mathsf{Y}_j \rangle_{L^2(\mu_0^j)} \to (\kappa_0\kappa_1)^{-1/2}. \tag{4.14}$$

Therefore, we have

$$\left\| \mathsf{Y}_j - \sqrt{\frac{\kappa_0}{\kappa_1}} X_j \right\|^2_{L^2(\mu_0^j)} = \|\mathsf{Y}_j\|^2_{L^2(\mu_0^j)} + \frac{\kappa_0}{\kappa_1} \|X_j\|^2_{L^2(\mu_0^j)} - 2\sqrt{\frac{\kappa_0}{\kappa_1}} \langle X_j, \mathsf{Y}_j \rangle_{L^2(\mu_0^j)} \to 0. \quad (4.15)$$

Step 2: Near-extremizers at the endpoints. We now use the limits of the vector fields to obtain near-extremizers at the two endpoints. Define $v_j = u_j \circ T_j^{-1}$. Since T_j is bi-Lipschitz and $(T_j)_\# \mu_0^j = \mu_1^j$, we have $v_j \in H^1(\mu_1^j)$ and $\int v_j \, \mathrm{d}\mu_1^j = 0$. The Sobolev chain rule and change of variables yield

$$(\nabla v_j) \circ T_j = X_j, \qquad \|v_j\|_{L^2(\mu_1^j)} = \|u_j\|_{L^2(\mu_0^j)}. \quad (4.16)$$

Testing the weak divergence equation in (4.10) with u_j, we obtain

$$\int g_j u_j \, \mathrm{d}\mu_0^j = \int X_j \cdot \nabla u_j \, \mathrm{d}\mu_0^j = \langle X_j, \mathsf{Y}_j \rangle_{L^2(\mu_0^j)}. \quad (4.17)$$

Together with (4.14), this gives

$$\left| \int g_j u_j \, \mathrm{d}\mu_0^j \right| = (\kappa_0 \kappa_1)^{-1/2} + o(1). \quad (4.18)$$

Since $\|g_j\|_{L^2(\mu_0^j)} = 1$, the endpoint Brascamp–Lieb inequalities and (4.16) yield

$$\begin{aligned} (\kappa_0\kappa_1)^{-1/2} + o(1) &\le \|u_j\|_{L^2(\mu_0^j)} \|g_j\|_{L^2(\mu_0^j)} = \|v_j\|_{L^2(\mu_1^j)} \\ &\le \min\left\{ \kappa_0^{-1/2} \|\mathsf{Y}_j\|_{L^2(\mu_0^j)}, \kappa_1^{-1/2} \|X_j\|_{L^2(\mu_0^j)} \right\} \\ &= (\kappa_0\kappa_1)^{-1/2} + o(1), \end{aligned} \quad (4.19)$$

which implies $\|u_j\|_{L^2(\mu_0^j)} = \|v_j\|_{L^2(\mu_1^j)} \to (\kappa_0\kappa_1)^{-1/2}$ and the endpoint Poincaré deficits

$$\delta_{0j} = \int |\nabla u_j|^2 \, \mathrm{d}\mu_0^j - \kappa_0 \int u_j^2 \, \mathrm{d}\mu_0^j \to 0, \qquad \delta_{1j} = \int |\nabla v_j|^2 \, \mathrm{d}\mu_1^j - \kappa_1 \int v_j^2 \, \mathrm{d}\mu_1^j \to 0. \quad (4.20)$$

Let $p_j = \int \nabla u_j \, \mathrm{d}\mu_0^j$ and $q_j = \int \nabla v_j \, \mathrm{d}\mu_1^j$. Since the regularized potentials have the same curvature lower bounds as the original endpoints, Lemma 4.1 gives

$$\nabla u_j - p_j \to 0 \quad \text{in } L^2(\mu_0^j), \qquad \nabla v_j - q_j \to 0 \quad \text{in } L^2(\mu_1^j). \quad (4.21)$$

Since $p_j = \int \mathsf{Y}_j \, \mathrm{d}\mu_0^j$ and $q_j = \int X_j \, \mathrm{d}\mu_0^j$, the orthogonal decompositions

$$\|\mathsf{Y}_j\|^2_{L^2(\mu_0^j)} = \|\mathsf{Y}_j - p_j\|^2_{L^2(\mu_0^j)} + |p_j|^2, \qquad \|X_j\|^2_{L^2(\mu_0^j)} = \|X_j - q_j\|^2_{L^2(\mu_0^j)} + |q_j|^2 \quad (4.22)$$

imply $|p_j| \to \kappa_1^{-1/2}$ and $|q_j| \to \kappa_0^{-1/2}$. Combining this with (4.15), we obtain

$$\left| p_j - \sqrt{\frac{\kappa_0}{\kappa_1}} q_j \right| \le \left\| \mathsf{Y}_j - \sqrt{\frac{\kappa_0}{\kappa_1}} X_j \right\|_{L^2(\mu_0^j)} \to 0. \quad (4.23)$$

Since $|\sqrt{\kappa_1}\, p_j| \to 1$ and $\sqrt{\kappa_1}\, p_j - \sqrt{\kappa_0}\, q_j \to 0$, after passing to a subsequence, there exists $e \in \mathbb{R}^n$ with $|e| = 1$ such that

$$\sqrt{\kappa_1}\, p_j \to e, \qquad \sqrt{\kappa_0}\, q_j \to e. \quad (4.24)$$

Step 3: Affine limits and Gaussian integration by parts. Let $m_i^j = \int x \, \mathrm{d}\mu_i^j$. By (4.21) and the endpoint Poincaré inequalities,

$$\left\| u_j - p_j \cdot (x - m_0^j) \right\|_{H^1(\mu_0^j)} \to 0, \qquad \left\| v_j - q_j \cdot (x - m_1^j) \right\|_{H^1(\mu_1^j)} \to 0. \tag{4.25}$$

For $i = 0, 1$ and $r, s \in H^1(\mu_i^j)$, consider the symmetric positive-semidefinite bilinear form

$$Q_i^{(j)}(r, s) = \int \nabla r \cdot \nabla s \, \mathrm{d}\mu_i^j - \kappa_i \operatorname{Cov}_{\mu_i^j}(r, s).$$

The Poincaré inequality makes $Q_i^{(j)}$ positive semidefinite, so it satisfies the Cauchy–Schwarz inequality. For fixed $\psi \in C_c^\infty(\mathbb{R}^n)$,

$$0 \le Q_i^{(j)}(\psi, \psi) \le \int |\nabla \psi|^2 \, \mathrm{d}\mu_i^j \le \|\nabla \psi\|_{L^\infty}^2 , \qquad \text{uniformly in } j, \tag{4.26}$$

which implies

$$|Q_0^{(j)}(u_j, \psi)|^2 \le \delta_{0j} Q_0^{(j)}(\psi, \psi) \to 0, \qquad |Q_1^{(j)}(v_j, \psi)|^2 \le \delta_{1j} Q_1^{(j)}(\psi, \psi) \to 0. \tag{4.27}$$

Let $\ell_{0j} = p_j \cdot (x - m_0^j)$. By (4.25),

$$|Q_0^{(j)}(u_j - \ell_{0j}, \psi)|^2 \le \|\nabla(u_j - \ell_{0j})\|_{L^2(\mu_0^j)}^2 \, Q_0^{(j)}(\psi, \psi) \to 0. \tag{4.28}$$

Hence we have

$$Q_0^{(j)}(\ell_{0j}, \psi) = Q_0^{(j)}(u_j, \psi) - Q_0^{(j)}(u_j - \ell_{0j}, \psi) \to 0, \tag{4.29}$$

and the analogous conclusion holds at the target. Expanding the source identity yields

$$Q_0^{(j)}\big(p_j \cdot (x - m_0^j), \psi\big) = p_j \cdot \left[\int \nabla \psi \, \mathrm{d}\mu_0^j - \kappa_0 \int (x - m_0^j) \psi \, \mathrm{d}\mu_0^j \right], \tag{4.30}$$

and the target identity is the same with q_j, μ_1^j, κ_1. Moreover, for fixed $\psi \in C_c^\infty(\mathbb{R}^n)$, the functions $\nabla\psi$, ψ, and $x\psi(x)$ are bounded and continuous. Thus every term in the brackets converges under weak convergence, while $m_i^j \to m_i$ follows from W_2 convergence. Combining with (4.24) and passing to the limit, we obtain

$$\int e \cdot \nabla \psi \, \mathrm{d}\mu_i = \kappa_i \int e \cdot (x - m_i) \psi(x) \, \mathrm{d}\mu_i, \qquad \psi \in C_c^\infty(\mathbb{R}^n), \quad i = 0, 1. \tag{4.31}$$

Step 4: Gaussian factorization at both endpoints. We finish the necessity by identifying the measures satisfying (4.31). For $i = 0, 1$, define the deterministic coordinate maps

$$s_i(x) = e \cdot (x - m_i), \qquad z_i(x) = P_{e^\perp}(x - m_i).$$

If $X_i \sim \mu_i$, then $S_i = s_i(X_i)$ and $Z_i = z_i(X_i)$ are the corresponding random coordinates. Their joint characteristic function is

$$\phi_i(\tau, \xi) = \mathbb{E}\, e^{\,\mathrm{i}(\tau S_i + \xi \cdot Z_i)} = \int e^{\,\mathrm{i}(\tau s_i(x) + \xi \cdot z_i(x))} \, \mathrm{d}\mu_i(x), \qquad \tau \in \mathbb{R}, \quad \xi \in e^\perp. \tag{4.32}$$

Choose a radial cut-off function $\chi \in C_c^\infty(\mathbb{R}^n)$ such that $0 \le \chi \le 1$, $\chi = 1$ on B_1, and $\operatorname{supp}\chi \subseteq B_2$. We may apply (4.31) to the real and imaginary parts of $\psi_R := \chi((x - m_i)/R)\, e^{\,\mathrm{i}(\tau s_i(x) + \xi \cdot z_i(x))}$ and then combine the resulting identities. Note that

$$e \cdot \nabla \psi_R(x) = \frac{1}{R} e \cdot \nabla \chi\left(\frac{x - m_i}{R} \right) e^{\,\mathrm{i}(\tau s_i(x) + \xi \cdot z_i(x))} + \mathrm{i}\tau \chi\left(\frac{x - m_i}{R} \right) e^{\,\mathrm{i}(\tau s_i(x) + \xi \cdot z_i(x))}. \tag{4.33}$$

Applying (4.31) to ψ_R and then passing to the limit, we obtain

$$\mathrm{i}\tau\phi_i(\tau,\xi) = \kappa_i \int s_i(x) e^{\,\mathrm{i}(\tau s_i(x)+\xi\cdot z_i(x))}\,\mathrm{d}\mu_i(x) = \kappa_i \mathrm{i}^{-1}\partial_\tau \phi_i(\tau,\xi), \tag{4.34}$$

which implies $\phi_i(\tau,\xi) = e^{-\tau^2/(2\kappa_i)}\phi_i(0,\xi)$. By uniqueness of characteristic functions, $S_i \sim N(0,\kappa_i^{-1})$ and is independent of Z_i. Applying this argument to $i = 0, 1$ proves (1.2) with the same direction e.

Step 5: The converse and equality at all times. Conversely, suppose that (1.2) holds. Let G_i denote the Gaussian factor in (1.2). Note that the quadratic cost is additive under $\mathbb{R}e \oplus e^\perp$. Hence the product of optimal couplings between (G_0, G_1) and (ν_0, ν_1) is optimal between the full endpoints. Since μ_0 is absolutely continuous, the quadratic optimal transport plan is unique and therefore coincides with this product plan. Thus the optimal transport splits along $\mathbb{R}e \oplus e^\perp$.

Let (X_0, X_1) have the optimal joint law and set $S_i = e \cdot (X_i - m_i)$ for $i = 0, 1$. Then S_0, S_1 are the centered Gaussian coordinates of the two marginals. Moreover, the optimal Gaussian coupling satisfies $S_1 = \sqrt{\kappa_0/\kappa_1}\, S_0$ almost surely. At time $t \in (0,1)$, let

$$\ell_t(x) = e \cdot (x - m_t), \qquad m_t = \int x\,\mathrm{d}\mu_t.$$

Set $X_t = (1-t)X_0 + tX_1$, so that $X_t \sim \mu_t$ and $m_t = (1-t)m_0 + tm_1$. Hence

$$\ell_t(X_t) = (1-t)S_0 + tS_1 = \left((1-t) + t\sqrt{\frac{\kappa_0}{\kappa_1}}\right) S_0. \tag{4.35}$$

Consequently, since $\mathrm{Var}(S_0) = \kappa_0^{-1}$ and $\nabla \ell_t = e$ with $|e| = 1$, we obtain

$$\mathrm{Var}_{\mu_t}(\ell_t) = \left(\frac{1-t}{\sqrt{\kappa_0}} + \frac{t}{\sqrt{\kappa_1}}\right)^2, \qquad \int |\nabla \ell_t|^2\,\mathrm{d}\mu_t = 1. \tag{4.36}$$

Although ℓ_t is not compactly supported, its standard cutoffs $\ell_{t,R}(x) = \chi(x/R)\ell_t(x)$, where $\chi \in C_c^\infty(\mathbb{R}^n)$ is radial, $\chi = 1$ on B_1, and $\operatorname{supp}\chi \subseteq B_2$, satisfy

$$\ell_{t,R} \longrightarrow \ell_t \qquad \text{in } H^1(\mu_t). \tag{4.37}$$

Indeed,

$$\nabla \ell_{t,R} = \chi(x/R)e + R^{-1}\ell_t(x)(\nabla\chi)(x/R), \tag{4.38}$$

and the convergence follows from $\mu_t \in \mathcal{P}_2(\mathbb{R}^n)$.

Therefore, by the definition of the Poincaré constant and the convergences in (4.37),

$$C_P(\mu_t) \geq \lim_{R\to\infty} \frac{\mathrm{Var}_{\mu_t}(\ell_{t,R})}{\int |\nabla \ell_{t,R}|^2\,\mathrm{d}\mu_t} = \left(\frac{1-t}{\sqrt{\kappa_0}} + \frac{t}{\sqrt{\kappa_1}}\right)^2. \tag{4.39}$$

Combining with Theorem 1.1 proves the result. Replacing t by any $s \in [0,1]$ in (4.37) and (4.39) proves equality at every time. □

The equality argument identifies all common Gaussian directions, not only a single one. This yields the following maximal factorization.

Proposition 4.2 (Maximal common Gaussian factor)**.** *Let μ_i be κ_i-strongly log-concave probability measures on $\mathbb{R}^n$, let $m_i = \int x \, \mathrm{d}\mu_i$, and define*

$$E = \left\{ v \in \mathbb{R}^n : \int v \cdot \nabla \psi \, \mathrm{d}\mu_i = \kappa_i \int v \cdot (x - m_i) \psi \, \mathrm{d}\mu_i, \quad \forall \, \psi \in C_c^\infty(\mathbb{R}^n), \quad i = 0, 1 \right\}. \tag{4.40}$$

Then E is a linear subspace. Moreover, under the orthogonal identification $\mathbb{R}^n = E \oplus E^\perp$, there exist $\nu_i \in \mathcal{P}(E^\perp)$ such that

$$\mu_i = N_E(P_E m_i, \kappa_i^{-1} I_E) \otimes \nu_i, \qquad i = 0, 1. \tag{4.41}$$

Here N_E denotes a Gaussian law on E. If a subspace $F \subseteq \mathbb{R}^n$ admits such a factorization for both endpoints, with respective covariances $\kappa_i^{-1} I_F$, then $F \subseteq E$. Thus E is the unique maximal common Gaussian subspace. Moreover, equality in Theorem 1.1 at one (equivalently, every) time $t \in (0,1)$ holds if and only if $E \neq \{0\}$.

Proof. The defining identities in (4.40) are linear in v, so E is a linear subspace. Fix $i \in \{0,1\}$ and set $s = P_E(x - m_i)$ and $z = P_{E^\perp}(x - m_i)$. For $\tau \in E$ and $\xi \in E^\perp$, let $\phi_i(\tau, \xi) = \int e^{\mathrm{i}(\tau \cdot s + \xi \cdot z)} \, \mathrm{d}\mu_i$. Applying (4.40) to cut-off approximations of the exponential, exactly as in the proof of Theorem 1.4, we obtain

$$v \cdot \nabla_\tau \phi_i(\tau, \xi) = -\kappa_i^{-1} (v \cdot \tau) \phi_i(\tau, \xi), \qquad \text{for every } v \in E, \tag{4.42}$$

which implies $\nabla_\tau \phi_i(\tau, \xi) = -\kappa_i^{-1} \tau \phi_i(\tau, \xi)$, and hence $\phi_i(\tau, \xi) = e^{-|\tau|^2/(2\kappa_i)} \phi_i(0, \xi)$. By uniqueness of characteristic functions, this factorization shows that $s \sim N_E(0, \kappa_i^{-1} I_E)$ and is independent of z; after recentering, this yields (4.41).

Conversely, if both endpoints split off Gaussian factors on a subspace F with covariances $\kappa_i^{-1} I_F$, then Gaussian integration by parts shows that every $v \in F$ satisfies (4.40). Hence $F \subseteq E$, so E is the unique maximal common Gaussian subspace. Finally, Theorem 1.4 shows that equality at an interior time is equivalent to $E \neq \{0\}$. □

Proposition 4.2 gives the following geometric description in the extended-valued case. For a closed convex set K, let $\mathrm{Lin}(K) := \{v : K + \mathbb{R}v = K\}$ denote its lineality space.

Corollary 4.3 (Conditioned Gaussian measures)**.** *Let $K_i \subseteq \mathbb{R}^n$ be convex Borel sets with nonempty interior, let $\kappa_i > 0$, and let μ_i be the probability measures with densities*

$$\mathrm{d}\mu_i(x) \propto e^{-\kappa_i |x|^2/2} \mathbf{1}_{K_i}(x) \, \mathrm{d}x, \qquad i = 0, 1.$$

Then their quadratic Wasserstein interpolation satisfies

$$C_P(\mu_t) \leq \left(\frac{1-t}{\sqrt{\kappa_0}} + \frac{t}{\sqrt{\kappa_1}} \right)^2, \qquad 0 \leq t \leq 1. \tag{4.43}$$

Equality holds for some $t \in (0,1)$ if and only if

$$\mathrm{Lin}(\bar{K}_0) \cap \mathrm{Lin}(\bar{K}_1) \neq \{0\}. \tag{4.44}$$

In that case equality holds for every $t \in [0,1]$. In particular, if either K_0 or K_1 is bounded, then

$$C_P(\mu_t) < \left(\frac{1-t}{\sqrt{\kappa_0}} + \frac{t}{\sqrt{\kappa_1}} \right)^2, \qquad \textit{for every } t \in (0,1). \tag{4.45}$$

Proof. Since the boundary of a convex set with nonempty interior has Lebesgue measure zero, replacing K_i by $\bar{K}_i$ does not change μ_i. We may assume that both sets are closed. Up to additive constants, the extended-valued endpoint potentials are $V_i(x) = \frac{\kappa_i}{2}|x|^2 + \iota_{K_i}(x)$, where ι_{K_i} is zero on K_i and $+\infty$ on K_i^c. Hence Theorem 1.1 implies the asserted inequality.

Suppose first that the two lineality spaces contain a common unit vector e. Denote $K_i' = K_i \cap e^\perp$. We claim that $K_i = \mathbb{R}e + K_i'$. Indeed, $K_i + \mathbb{R}e = K_i$, and subtracting the e-component of any point of K_i leaves a point of K_i'. The Gaussian density therefore factorizes under $\mathbb{R}^n = \mathbb{R}e \oplus e^\perp$, and hence $\mu_i = N(0, \kappa_i^{-1}) \otimes \nu_i$, where ν_i is the corresponding conditioned Gaussian measure on K_i'. The converse part of Theorem 1.4 yields equality at every time.

Conversely, by Theorem 1.4, equality at an interior time gives a common Gaussian line factor. Since the density of μ_i is positive on $\operatorname{int} K_i$ and $K_i = \operatorname{cl}(\operatorname{int} K_i)$, its support is K_i. On the other hand, the support of the product in (1.2) has the form $\mathbb{R}e + \operatorname{supp} \nu_i$. It is invariant under translation by $\mathbb{R}e$, and therefore $e \in \operatorname{Lin}(K_i)$ for both endpoints. Finally, a bounded convex set has trivial lineality space, which proves the strict inequality in the last assertion. □

5 Gaussian entropy and functional inequalities

We now apply the Poincaré interpolation theorem to the second variation of Gaussian relative entropy. Throughout this section, γ denotes the standard Gaussian measure, and $\mathcal{P}_2(\mathbb{R}^n)$ denotes the probability measures on $\mathbb{R}^n$ with finite second moment. We write

$$L = \Delta - x \cdot \nabla, \qquad \Gamma_2(u) = \|\operatorname{Hess} u\|_{\mathrm{HS}}^2 + |\nabla u|^2$$

for the Ornstein–Uhlenbeck generator and its iterated carré du champ, respectively. We also write $P_s = e^{sL}$ for the associated Markov semigroup on functions and P_s^* for its dual action on probability measures.

All the entropies appearing below are finite. Indeed, relative to γ, a 1-strongly log-concave endpoint has density e^{-W} with W convex, possibly extended-valued. Since W is bounded below by an affine function, its negative part grows at most linearly, while $se^{-s} \le e^{-1}$ for $s \ge 0$ controls its positive part. Thus $\operatorname{Ent}_\gamma(\mu_i) < \infty$. Moreover,

$$\operatorname{Ent}_\gamma(\mu) = \operatorname{Ent}_{\mathcal{L}^n}(\mu) + \frac{1}{2}\int |x|^2 \, \mathrm{d}\mu + \frac{n}{2}\log(2\pi).$$

The second moment remains finite along the Wasserstein geodesic, and $\operatorname{Ent}_{\mathcal{L}^n}$ is displacement convex [41]. Hence $\operatorname{Ent}_\gamma(\mu_t) < \infty$ for every $t \in [0,1]$.

When the mean is zero, the estimate below follows the argument in [3, Theorem 4.6], which subtracts a quadratic polynomial. In Lemma 5.1, we separate the translation component. This removes the zero-average and symmetry assumptions. The local-to-global comparison and the subsequent functional estimates follow the arguments used by Aishwarya and Rotem [3, Sections 4–5] and by Aishwarya and Li [2, proof of Theorem 1.7] to derive functional inequalities from entropy. We prove the changes required by the translation component and cite each unchanged step at the point where it is used.

Lemma 5.1 (Γ_2 estimate with the mean separated)**.** *Let $\mu \in \mathcal{P}_2(\mathbb{R}^n)$ satisfy $C_P(\mu) \le 1$, and let $u \in C^\infty(\mathbb{R}^n)$ satisfy $\nabla u, \operatorname{Hess} u \in L^2(\mu)$. Denote*

$$m = \int x \, \mathrm{d}\mu, \qquad a = \int \nabla u \, \mathrm{d}\mu, \qquad \ell = \int Lu \, \mathrm{d}\mu.$$

Then

$$\int \Gamma_2(u)\,\mathrm{d}\mu \geq 2\int |\nabla u|^2\,\mathrm{d}\mu - |a|^2 + \frac{(\ell + m\cdot a)^2}{n} + \frac{(\ell + m\cdot a)^2}{n^2}\int |x-m|^2\,\mathrm{d}\mu. \tag{5.1}$$

Proof. The proof follows [3, Theorem 4.6]; we give the modifications involving m and a. First suppose that $a = 0$. Set $\lambda = \ell/n$ and define $v(x) = u(x) - \frac{\lambda}{2}|x-m|^2$. Then $\int \nabla v\,\mathrm{d}\mu = 0$. The derivative $\partial_j v$ need not be compactly supported. We therefore apply the Poincaré inequality to $\eta_R \partial_j v$, where η_R is a standard radial cut-off, and then let $R \to \infty$. This is the same cut-off argument used in the proof of Theorem 1.4. The error terms vanish since $\nabla v, \operatorname{Hess} v \in L^2(\mu)$. Summing over $j = 1, \dots, n$, gives

$$\int \|\operatorname{Hess} v\|_{\mathrm{HS}}^2\,\mathrm{d}\mu \geq \int |\nabla v|^2\,\mathrm{d}\mu. \tag{5.2}$$

Expanding (5.2) and using $\nabla v = \nabla u - \lambda(x-m)$ and $\operatorname{Hess} v = \operatorname{Hess} u - \lambda I$, we obtain

$$\begin{aligned}\int \|\operatorname{Hess} u\|_{\mathrm{HS}}^2\,\mathrm{d}\mu - 2\lambda\int \Delta u\,\mathrm{d}\mu + n\lambda^2 \geq \int |\nabla u|^2\,\mathrm{d}\mu \\ - 2\lambda\int (x-m)\cdot\nabla u\,\mathrm{d}\mu + \lambda^2\int |x-m|^2\,\mathrm{d}\mu.\end{aligned} \tag{5.3}$$

Since $a = 0$, $\int \big(\Delta u - (x-m)\cdot\nabla u\big)\,\mathrm{d}\mu = \ell$. Thus,

$$\begin{aligned}\int \|\operatorname{Hess} u\|_{\mathrm{HS}}^2\,\mathrm{d}\mu &\geq \int |\nabla u|^2\,\mathrm{d}\mu + 2\lambda\ell - n\lambda^2 + \lambda^2\int |x-m|^2\,\mathrm{d}\mu \\ &= \int |\nabla u|^2\,\mathrm{d}\mu + \frac{\ell^2}{n} + \frac{\ell^2}{n^2}\int |x-m|^2\,\mathrm{d}\mu,\end{aligned} \tag{5.4}$$

which implies (5.1) when $a = 0$.

For general a, apply the case $a = 0$ to $w(x) = u(x) - a\cdot x$. Then

$$\int \nabla w\,\mathrm{d}\mu = 0, \qquad \int Lw\,\mathrm{d}\mu = \ell + m\cdot a, \qquad \int |\nabla w|^2\,\mathrm{d}\mu = \int |\nabla u|^2\,\mathrm{d}\mu - |a|^2. \tag{5.5}$$

Since $\operatorname{Hess} w = \operatorname{Hess} u$,

$$\int \Gamma_2(w)\,\mathrm{d}\mu = \int \Gamma_2(u)\,\mathrm{d}\mu - |a|^2. \tag{5.6}$$

Applying the estimate for the case $a = 0$ to w then yields (5.1). □

Proof of Theorem 1.5. **Step 1: Entropy variations.** First assume that the endpoint potentials are smooth with two-sided Hessian bounds. Let $\nabla\theta_t$ be the velocity of the constant-speed geodesic, so

$$\partial_t \mu_t + \operatorname{div}(\mu_t \nabla\theta_t) = 0, \qquad \partial_t\theta_t + \frac{1}{2}|\nabla\theta_t|^2 = 0. \tag{5.7}$$

Write $\mu_t = \rho_t\gamma$ and set $E_t := \operatorname{Ent}_\gamma(\mu_t) = \int \rho_t \log\rho_t\,\mathrm{d}\gamma$. The first- and second-variation formulas in [3, Lemma 4.3] yield

$$E_t' = \int \nabla\log\rho_t\cdot\nabla\theta_t\,\mathrm{d}\mu_t = \int \nabla\rho_t\cdot\nabla\theta_t\,\mathrm{d}\gamma = -\int L\theta_t\,\mathrm{d}\mu_t, \qquad E_t'' = \int \Gamma_2(\theta_t)\,\mathrm{d}\mu_t, \tag{5.8}$$

where the assumed regularity and Gaussian tails justify the differentiations and integrations by parts above via standard radial cutoffs.

Step 2: Separating the translation component. Set $m_t = \int x \, \mathrm{d}\mu_t$ and $a = m_1 - m_0$. The continuity equation and the constant-speed property of the geodesic imply

$$m_t = (1-t)m_0 + tm_1, \qquad \int \nabla\theta_t \, \mathrm{d}\mu_t = m_t' = a, \qquad \int |\nabla\theta_t|^2 \, \mathrm{d}\mu_t = W_2^2(\mu_0, \mu_1). \tag{5.9}$$

Here a is precisely the translation velocity. Set $W_{\mathrm{c}}^2 = W_2^2(\mu_0, \mu_1) - |a|^2$. Since both endpoints are 1-strongly log-concave, Theorem 1.1 gives $C_P(\mu_t) \le 1$. By (5.8) and (5.9), the function $\mathcal{D}_t = E_t - |m_t|^2/2$ satisfies

$$\mathcal{D}_t' = E_t' - m_t \cdot a, \qquad \mathcal{D}_t'' = E_t'' - |a|^2, \tag{5.10}$$

and

$$\int L\theta_t \, \mathrm{d}\mu_t + m_t \cdot a = -\mathcal{D}_t', \qquad \int |\nabla\theta_t|^2 \, \mathrm{d}\mu_t - |a|^2 = W_{\mathrm{c}}^2. \tag{5.11}$$

Applying Lemma 5.1 to θ_t, we obtain

$$\mathcal{D}_t'' \ge 2W_{\mathrm{c}}^2 + \frac{(\mathcal{D}_t')^2}{n} + \frac{(\mathcal{D}_t')^2}{n^2} \int |x - m_t|^2 \, \mathrm{d}\mu_t, \tag{5.12}$$

which implies (1.4).

Step 3: Global comparison. For $i = 0, 1$, let $\bar{\mu}_i = (x \mapsto x - m_i)_{\#}\mu_i$ be the centered translate of μ_i. For every coupling π of the endpoints,

$$\int |x - y|^2 \, \mathrm{d}\pi = |m_1 - m_0|^2 + \int |(x - m_0) - (y - m_1)|^2 \, \mathrm{d}\pi, \tag{5.13}$$

which implies $W_{\mathrm{c}} = W_2(\bar{\mu}_0, \bar{\mu}_1)$. Translating the optimal transport plan also yields the centered interpolation $\bar{\mu}_t = (x \mapsto x - m_t)_{\#}\mu_t$. The Brascamp–Lieb inequality yields $\mathrm{Cov}(\bar{\mu}_i) \preceq I$. If $\bar{X}_0$ and $\bar{X}_1$ are independent with these laws, then

$$W_{\mathrm{c}}^2 \le \mathbb{E}|\bar{X}_0 - \bar{X}_1|^2 = \operatorname{tr}\mathrm{Cov}(\bar{\mu}_0) + \operatorname{tr}\mathrm{Cov}(\bar{\mu}_1) \le 2n. \tag{5.14}$$

Thus $\omega = \sqrt{2/n}\, W_{\mathrm{c}} \le 2 < \pi$. Setting $U_t = e^{-\mathcal{D}_t/n}$, (5.12) gives

$$U_t'' + \omega^2 U_t \le 0. \tag{5.15}$$

The one-dimensional comparison principle then yields (1.3). This is the standard local-to-global comparison for $(2, n)$-convexity; see [3, Lemma 5.3] and [20, Lemma 2.2].

Step 4: Equality in the smooth case. Keeping the last term in (5.12) yields

$$-U_t'' - \omega^2 U_t \ge \frac{U_t (\mathcal{D}_t')^2}{n^3} \int |x - m_t|^2 \, \mathrm{d}\mu_t. \tag{5.16}$$

For $0 \le \omega < \pi$, the Dirichlet Green kernel G_ω of $-\frac{\mathrm{d}^2}{\mathrm{d}t^2} - \omega^2$ on $(0, 1)$ is given by

$$G_\omega(t, s) = \frac{\sin(\omega t)\sin(\omega(1-s))}{\omega \sin\omega}, \qquad 0 < t \le s < 1, \tag{5.17}$$

and $G_\omega(t, s) = G_\omega(s, t)$ when $s < t$. At $\omega = 0$, the right-hand side is understood by continuity and equals $t(1 - s)$. In particular, $G_\omega(t, s) > 0$ for all $t, s \in (0, 1)$.

Since the sine interpolation below solves the homogeneous equation $v'' + \omega^2 v = 0$ and agrees with U at $t = 0, 1$, the Green representation yields, for every $t_0 \in (0, 1)$,

$$U_{t_0} - \frac{\sin((1-t_0)\omega)}{\sin\omega} U_0 - \frac{\sin(t_0\omega)}{\sin\omega} U_1 = \int_0^1 G_\omega(t_0, s)\big(-U_s'' - \omega^2 U_s\big) \, \mathrm{d}s \ge 0. \tag{5.18}$$

Suppose that equality holds at some $t_0 \in (0,1)$. Since $G_\omega(t_0, s) > 0$,

$$-U_s'' - \omega^2 U_s = 0, \qquad \text{for almost every } s \in (0,1). \tag{5.19}$$

Since μ_s is not a point mass, $\int |x - m_s|^2 \, \mathrm{d}\mu_s > 0$. Combining with (5.16) yields $\mathcal{D}_s' = 0$ for almost every $s \in (0,1)$. Under the present smoothness assumptions, $\mathcal{D}'$ is continuous, and hence $\mathcal{D}_s' = 0$ for every $s \in (0,1)$. It follows from (5.12) that $W_c = 0$. Hence μ_1 is a translate of μ_0. Conversely, if μ_1 is a translate of μ_0, then the centered geodesic is constant. Hence $W_c = 0$ and $\mathcal{D}_t$ is constant, so equality holds for every $t \in [0,1]$.

Step 5: Regularization. It remains to remove the smoothness assumptions. Denote $s_\varepsilon = \frac{1}{2}\log(1+\varepsilon)$ and define $\bar{\mu}_i^\varepsilon = P_{s_\varepsilon}^* \bar{\mu}_i$. By Lemma 3.2, the measures $\bar{\mu}_i^\varepsilon$ are centered, smooth, 1-strongly log-concave and $\bar{\mu}_i^\varepsilon \to \bar{\mu}_i$ in W_2. Let $(\bar{\mu}_t^\varepsilon)_{t\in[0,1]}$ denote their quadratic Wasserstein geodesic, set $W_\varepsilon := W_2(\bar{\mu}_0^\varepsilon, \bar{\mu}_1^\varepsilon)$ and $\omega_\varepsilon := \sqrt{2/n}\, W_\varepsilon$.

Since $P_{s_\varepsilon}^* \gamma = \gamma$, the data processing inequality yields $\mathrm{Ent}_\gamma(\bar{\mu}_i^\varepsilon) \le \mathrm{Ent}_\gamma(\bar{\mu}_i)$. On the other hand, lower semicontinuity of relative entropy gives $\mathrm{Ent}_\gamma(\bar{\mu}_i) \le \liminf_{\varepsilon\to 0} \mathrm{Ent}_\gamma(\bar{\mu}_i^\varepsilon)$; this follows directly from the Donsker–Varadhan variational formula [17, Section 2]. Hence $\mathrm{Ent}_\gamma(\bar{\mu}_i^\varepsilon) \longrightarrow \mathrm{Ent}_\gamma(\bar{\mu}_i)$. Stability of optimal transport plans, as in the proof of Theorem 1.1, gives $\bar{\mu}_t^\varepsilon \to \bar{\mu}_t$ in W_2 and $W_\varepsilon \to W_c$. Apply the smooth inequality to $(\bar{\mu}_t^\varepsilon)_{t\in[0,1]}$. The endpoint entropies converge, and entropy is lower semicontinuous at time t. We may therefore pass to the limit and obtain the same inequality for $\mathrm{Ent}_\gamma(\bar{\mu}_t)$. Since

$$\mathrm{Ent}_\gamma(\bar{\mu}_t) = \mathrm{Ent}_\gamma(\mu_t) - \frac{1}{2}|m_t|^2 = \mathcal{D}_t, \tag{5.20}$$

this is precisely (1.3).

For the equality case, set

$$U_s^\varepsilon := e^{-\mathrm{Ent}_\gamma(\bar{\mu}_s^\varepsilon)/n}$$

and let $S_\varepsilon(t_0)$ denote the sine interpolation of U_0^ε and U_1^ε with frequency ω_ε. Thus the gap for the regularized geodesic is $\Delta_\varepsilon(t_0) = U_{t_0}^\varepsilon - S_\varepsilon(t_0) \ge 0$. The convergence of the endpoint entropies and of ω_ε gives $S_\varepsilon(t_0) \to S(t_0)$, the corresponding sine interpolation for the limiting endpoints. Lower semicontinuity at the interior time gives

$$\mathrm{Ent}_\gamma(\bar{\mu}_{t_0}) \le \liminf_{\varepsilon\to 0} \mathrm{Ent}_\gamma(\bar{\mu}_{t_0}^\varepsilon), \qquad U_{t_0} \ge \limsup_{\varepsilon\to 0} U_{t_0}^\varepsilon. \tag{5.21}$$

If equality holds at $t_0 \in (0,1)$ for the limiting geodesic, then $U_{t_0} = S(t_0)$, and consequently

$$0 \le \limsup_{\varepsilon\to 0} \Delta_\varepsilon(t_0) \le U_{t_0} - S(t_0) = 0. \tag{5.22}$$

Hence $\Delta_\varepsilon(t_0) \to 0$. Let $M_s^\varepsilon = \int |x|^2 \, \mathrm{d}\bar{\mu}_s^\varepsilon$. Since $M_i^\varepsilon \to \int |x|^2 \, \mathrm{d}\bar{\mu}_i > 0$ for $i = 0, 1$, every sufficiently small $\varepsilon > 0$ satisfies

$$M_s^\varepsilon = (1-s)M_0^\varepsilon + sM_1^\varepsilon - s(1-s)W_\varepsilon^2 \ge (1-s)^2 M_0^\varepsilon + s^2 M_1^\varepsilon > 0, \qquad \frac{1}{3} \le s \le \frac{2}{3}. \tag{5.23}$$

Combining (5.16) and (5.18), we obtain

$$\begin{aligned}
\Delta_\varepsilon(t_0) &\ge \frac{1}{n^3} \int_0^1 G_{\omega_\varepsilon}(t_0, s) e^{-\mathrm{Ent}_\gamma(\bar{\mu}_s^\varepsilon)/n} M_s^\varepsilon \left| \frac{\mathrm{d}}{\mathrm{d}s} \mathrm{Ent}_\gamma(\bar{\mu}_s^\varepsilon) \right|^2 \mathrm{d}s \\
&\ge C \int_{1/3}^{2/3} \left| \frac{\mathrm{d}}{\mathrm{d}s} \mathrm{Ent}_\gamma(\bar{\mu}_s^\varepsilon) \right|^2 \mathrm{d}s \\
&\ge \frac{3C}{2} \int_{1/3}^{2/3} \int_{1/3}^{2/3} \left| \frac{\mathrm{d}}{\mathrm{d}s} \mathrm{Ent}_\gamma(\bar{\mu}_s^\varepsilon) - \frac{\mathrm{d}}{\mathrm{d}r} \mathrm{Ent}_\gamma(\bar{\mu}_r^\varepsilon) \right|^2 \mathrm{d}s \, \mathrm{d}r \overset{(5.12)}{\ge} \frac{C}{81} W_\varepsilon^4,
\end{aligned} \tag{5.24}$$

where C is independent of all sufficiently small ε. Since $\Delta_\varepsilon(t_0) \to 0$ and $W_\varepsilon \to W_c$, we obtain $W_c = 0$ and then μ_1 is a translate of μ_0. The converse proved above does not require smoothness.

Finally, if $m_0 = m_1$, then $m_t = m_0$, $W_c = W_2(\mu_0, \mu_1)$, and $\mathcal{D}_t = \mathrm{Ent}_\gamma(\mu_t) - \frac{1}{2}|m_0|^2$. Hence the same conclusions hold with $\mathrm{Ent}_\gamma(\mu_t)$ in place of $\mathcal{D}_t$. In the equality case, the translation vector must vanish, and hence $\mu_0 = \mu_1$. □

We next turn to the functional application. For a fixed $t \in (0,1)$, $a, b \geq 0$, and $p \in [0, \infty]$, define the weighted power mean

$$M_p^t(a,b) = \begin{cases} ((1-t)a^p + tb^p)^{1/p}, & ab > 0, \\ 0, & ab = 0, \end{cases}$$

with the usual limiting interpretations at $p = 0$ and $p = \infty$.

Theorem 5.2 (Gaussian BBL inequality with barycenter terms)**.** *Fix $t \in (0,1)$ and $p \in [0, \infty]$. Let $f, g, h \geq 0$ be γ-integrable, with f and g log-concave, and set $A = \int f \, \mathrm{d}\gamma > 0$ and $B = \int g \, \mathrm{d}\gamma > 0$. Assume that*

$$h((1-t)x + ty) \geq M_p^t(f(x), g(y)) \qquad \text{for every } x, y \in \mathbb{R}^n.$$

Define $\mathrm{d}\mu_f = \frac{f}{A} \mathrm{d}\gamma$ and $\mathrm{d}\mu_g = \frac{g}{B} \mathrm{d}\gamma$, and let b_f, b_g be the barycenters of μ_f, μ_g, respectively. Denote $b_t = (1-t)b_f + tb_g$ and $r = p/(1+np)$, with $r = 1/n$ when $p = \infty$. Then

$$\int h \, \mathrm{d}\gamma \geq e^{-|b_t|^2/2} M_r^t\Big(Ae^{|b_f|^2/2}, Be^{|b_g|^2/2}\Big). \tag{5.25}$$

In particular, if $b_f = b_g$, then

$$\int h \, \mathrm{d}\gamma \geq M_r^t(A, B). \tag{5.26}$$

Proof. We modify the argument of [2, Theorem 1.7] in two places to account for nonzero barycenters. Set $F = \log f$, $G = \log g$ and $H = \log h$ on their respective positivity sets. If (X, Y) is an optimal coupling of μ_f and μ_g, and μ_t is the law of $(1-t)X + tY$, then Jensen's inequality and Donsker–Varadhan duality yield

$$\int h \, \mathrm{d}\gamma \geq M_p^t\Big(e^{\int F \, \mathrm{d}\mu_f}, e^{\int G \, \mathrm{d}\mu_g}\Big) e^{-\mathrm{Ent}_\gamma(\mu_t)}. \tag{5.27}$$

Truncating the logarithms in the Donsker–Varadhan variational formula [17, Section 2] covers vanishing or unbounded functions, and the cases $p = 0, \infty$ follow by passage to the limit.

The measures μ_f and μ_g are 1-strongly log-concave, and the sine coefficients in (1.3) dominate their affine counterparts. Therefore, Theorem 1.5 gives

$$e^{-\mathcal{D}(\mu_t)/n} \geq (1-t)e^{-\mathcal{D}(\mu_f)/n} + te^{-\mathcal{D}(\mu_g)/n}, \tag{5.28}$$

which implies

$$e^{-\mathcal{D}(\mu_t)} \geq M_{1/n}^t\Big(e^{-\mathcal{D}(\mu_f)}, e^{-\mathcal{D}(\mu_g)}\Big). \tag{5.29}$$

We use the following two-point form of the weighted Hölder inequality:

$$M_p^t(a,b) M_{1/n}^t(c,d) \geq M_r^t(ac, bd), \qquad r = p/(1+np), \tag{5.30}$$

which is the weighted Hölder inequality with $1/r = 1/p + n$; the cases $p = 0$ and $p = \infty$ follow by continuity. Since the barycenter of μ_t is b_t, combining (5.27), (5.29), and (5.30) gives

$$\int h \, \mathrm{d}\gamma \geq e^{-|b_t|^2/2} M_r^t \Big(e^{\int F \, \mathrm{d}\mu_f - \mathcal{D}(\mu_f)}, e^{\int G \, \mathrm{d}\mu_g - \mathcal{D}(\mu_g)} \Big). \tag{5.31}$$

Since $\mathrm{d}\mu_f = e^F A^{-1} \, \mathrm{d}\gamma$ and $\mathrm{d}\mu_g = e^G B^{-1} \, \mathrm{d}\gamma$, we have

$$e^{\int F \, \mathrm{d}\mu_f - \mathcal{D}(\mu_f)} = A e^{|b_f|^2/2}, \qquad e^{\int G \, \mathrm{d}\mu_g - \mathcal{D}(\mu_g)} = B e^{|b_g|^2/2}, \tag{5.32}$$

which proves (5.25). □

Corollary 5.3 (Gaussian BBL inequality with sine coefficients)**.** *Under the assumptions of Theorem 5.2, denote*

$$\omega = \sqrt{2/n} \left(W_2^2(\mu_f, \mu_g) - |b_f - b_g|^2 \right)^{1/2},$$

and

$$\alpha = \frac{\sin((1-t)\omega)}{\sin \omega}, \qquad \beta = \frac{\sin(t\omega)}{\sin \omega},$$

with $\alpha = 1 - t$ and $\beta = t$ when $\omega = 0$. Then

$$\int h \, \mathrm{d}\gamma \geq e^{-|b_t|^2/2} M_r^t \left(A e^{|b_f|^2/2} \left(\frac{\alpha}{1-t} \right)^n, B e^{|b_g|^2/2} \left(\frac{\beta}{t} \right)^n \right). \tag{5.33}$$

Proof. The result follows by retaining the sine coefficients in (1.3) and applying (5.30) as in the proof of Theorem 5.2. □

Proof of Corollary 1.6. For $0 < t < 1$, apply Theorem 5.2 with

$$f = \mathbf{1}_{K_0}, \qquad g = \mathbf{1}_{K_1}, \qquad h = \mathbf{1}_{K_t}, \qquad p = \infty. \tag{5.34}$$

The pointwise assumption follows from the convention $M_p^t(a, b) = 0$ when $ab = 0$. Taking the n-th root of (5.25) yields (1.5).

For the equality case, apply Corollary 5.3 to the same indicator functions with $p = \infty$. Taking the n-th root gives the stronger estimate with sine coefficients

$$e^{|b_t|^2/(2n)} \gamma(K_t)^{1/n} \geq \alpha A_0 + \beta A_1, \qquad A_i = e^{|b_{K_i}|^2/(2n)} \gamma(K_i)^{1/n} > 0. \tag{5.35}$$

If $\omega > 0$, then $\alpha > 1 - t$ and $\beta > t$. Thus equality in (1.5) forces $\omega = 0$, and hence γ_{K_1} is a translate of γ_{K_0}. Set $a = b_{K_1} - b_{K_0}$. Comparing their densities with respect to Lebesgue measure yields $a \cdot x = \text{const}$ almost everywhere on the nonempty open set $\operatorname{int} K_1$, which forces $a = 0$. Thus $\gamma_{K_0} = \gamma_{K_1}$, and equality of their supports yields $\bar{K}_0 = \bar{K}_1$.

Conversely, if $\bar{K}_0 = \bar{K}_1$, then K_0, K_1, and K_t agree up to Gaussian-null boundaries, so equality holds for every $t \in [0, 1]$. □

Corollary 5.4 (Gaussian Brunn–Minkowski inequality with sine coefficients)**.** *Under the assumptions of Corollary 1.6, fix $t \in (0, 1)$ and denote*

$$\omega = \sqrt{2/n} \left(W_2^2(\gamma_{K_0}, \gamma_{K_1}) - |b_{K_1} - b_{K_0}|^2 \right)^{1/2},$$

and

$$\alpha = \frac{\sin((1-t)\omega)}{\sin \omega}, \qquad \beta = \frac{\sin(t\omega)}{\sin \omega},$$

with $\alpha = 1 - t$ and $\beta = t$ when $\omega = 0$. Then

$$e^{|b_t|^2/(2n)}\gamma(K_t)^{1/n} \geq \alpha e^{|b_{K_0}|^2/(2n)}\gamma(K_0)^{1/n} + \beta e^{|b_{K_1}|^2/(2n)}\gamma(K_1)^{1/n}. \tag{5.36}$$

Moreover, equality in (5.36) at some $t \in (0,1)$ holds if and only if $\bar{K}_0 = \bar{K}_1$. In this case, equality holds for every $t \in [0,1]$.

Proof. The result follows by applying Corollary 5.3 with the indicator functions used in the proof of Corollary 1.6, setting $p = \infty$, and taking the n-th root of (5.33).

For the equality case, let $(\mu_s)_{s\in[0,1]}$ be the quadratic Wasserstein geodesic from γ_{K_0} to γ_{K_1}. Since $\mu_t(K_t) = 1$, nonnegativity of relative entropy yields $\mathrm{Ent}_\gamma(\mu_t) \geq -\log \gamma(K_t)$. Together with Theorem 1.5, this gives

$$e^{|b_t|^2/(2n)}\gamma(K_t)^{1/n} \geq e^{-\mathcal{D}(\mu_t)/n} \geq \alpha A_0 + \beta A_1, \qquad A_i = e^{|b_{K_i}|^2/(2n)}\gamma(K_i)^{1/n} > 0. \tag{5.37}$$

Hence equality in (5.36) forces equality in (1.3), which implies γ_{K_1} is a translate of γ_{K_0}. The final density comparison and the converse are the same as in the proof of Corollary 1.6. □

We conclude with the functional inequalities obtained by joining a measure to equilibrium. This is the HWI principle of Otto and Villani [44], now applied with the centered entropy estimate. The even versions of the resulting HWI, logarithmic Sobolev, and Talagrand inequalities appear in [3, Section 5]; the centering argument in the proof of Corollary 5.5 removes the evenness assumption. If $\mathrm{d}\mu = \rho\,\mathrm{d}\gamma$, denote

$$\mathrm{I}_\gamma(\mu) = \int |\nabla \log \rho|^2 \,\mathrm{d}\mu$$

for relative Fisher information, with the usual value $+\infty$ outside its form domain.

Corollary 5.5 (Gaussian functional inequalities after centering)**.** *Let μ be 1-strongly log-concave, let $m = \int x\,\mathrm{d}\mu$, and denote*

$$\mathcal{D} = \mathrm{Ent}_\gamma(\mu) - \frac{1}{2}|m|^2, \quad \mathcal{I} = \mathrm{I}_\gamma(\mu) - |m|^2, \quad \mathcal{W}^2 = W_2^2(\mu,\gamma) - |m|^2.$$

Then $\mathcal{D} \geq 0$, $\mathcal{I} \geq 0$, and $0 \leq \mathcal{W}^2 \leq n$. Moreover,

$$e^{\mathcal{D}/n} \leq \cos\left(\sqrt{\frac{2}{n}}\,\mathcal{W}\right) + \frac{\sin\left(\sqrt{2/n}\,\mathcal{W}\right)}{\sqrt{2n}}\sqrt{\mathcal{I}}, \tag{5.38}$$

$$4\mathcal{D} \leq 2n\left(e^{2\mathcal{D}/n} - 1\right) \leq \mathcal{I}, \tag{5.39}$$

and

$$\mathcal{W}^2 \leq -n \log\cos\left(\sqrt{\frac{2}{n}}\,\mathcal{W}\right) \leq \mathcal{D}. \tag{5.40}$$

Moreover, for every $s \geq 0$,

$$\mathrm{Ent}_\gamma(P_s^*\mu) \leq e^{-4s}\left(\mathrm{Ent}_\gamma(\mu) - \frac{1}{2}|m|^2\right) + \frac{1}{2}e^{-2s}|m|^2. \tag{5.41}$$

Proof. **Step 1: Centering.** Let $\bar{\mu} = (x \mapsto x - m)_{\#}\mu$ be the centered translate of μ. Set $\mathrm{d}\mu = \rho\,\mathrm{d}\gamma$ and $\bar{\rho} = \mathrm{d}\bar{\mu}/\mathrm{d}\gamma$. Gaussian integration by parts and a change of variables give

$$\int \nabla \log \rho \,\mathrm{d}\mu = \int \nabla \rho \,\mathrm{d}\gamma = m, \qquad \log \bar{\rho}(x) = \log \rho(x+m) - m \cdot x - \frac{|m|^2}{2}, \tag{5.42}$$

where the Gaussian integration by parts in the first identity is justified by standard radial cutoffs whenever $\mathrm{I}_\gamma(\mu) < \infty$. If $\mathrm{I}_\gamma(\mu) = +\infty$, the identity for the translated density implies $\mathrm{I}_\gamma(\bar{\mu}) = +\infty$, so the Fisher-information identity below holds in the extended sense. Combining this with the orthogonal decomposition of the quadratic transport cost, we obtain

$$\mathcal{D} = \mathrm{Ent}_\gamma(\bar{\mu}), \qquad \mathcal{I} = \mathrm{I}_\gamma(\bar{\mu}), \qquad \mathcal{W} = W_2(\bar{\mu}, \gamma), \tag{5.43}$$

which yields the asserted nonnegativity. Hence it is sufficient to assume $m = 0$.

Set $\omega = \sqrt{2/n}\,\mathcal{W}$. In the smooth case, let T be the Brenier map from γ to $\bar{\mu}$. Since both measures are centered, $\int (T(x) - x)\,\mathrm{d}\gamma(x) = 0$, while Caffarelli's contraction theorem [9] yields $0 \preceq DT \preceq I$. Applying the Gaussian Poincaré inequality to each component of $T - I$ and summing over the components, we obtain

$$\mathcal{W}^2 = \int |T(x) - x|^2 \,\mathrm{d}\gamma(x) \le \int \|DT - I\|_{\mathrm{HS}}^2 \,\mathrm{d}\gamma \le n. \tag{5.44}$$

The general case follows by Lemma 3.2. In particular, $0 \le \omega \le \sqrt{2} < \pi/2$.

Step 2: HWI, logarithmic Sobolev, and Talagrand estimates. First assume that $\bar{\mu}$ has a smooth potential with two-sided Hessian bounds. Apply Theorem 1.5 to the geodesic from $\bar{\mu}$ to γ, and write $E_t = \mathrm{Ent}_\gamma(\mu_t)$. Differentiating the finite-time inequality at the first endpoint and using $-E_0' \le \sqrt{\mathrm{I}_\gamma(\bar{\mu})}\, W_2(\bar{\mu}, \gamma)$ yields (5.38), as in [3, Theorem 5.5]. Squaring this inequality and arguing as in [3, Corollary 5.6] yields (5.39). Reversing the geodesic and differentiating at the Gaussian endpoint, where the Fisher information vanishes, yields $e^{-\mathcal{D}/n} \le \cos\omega$. Together with $-\log\cos\omega \ge \omega^2/2$, this yields (5.40), as in [3, Corollary 5.11]. Once Theorem 1.5 has replaced the even entropy estimate used there, none of these arguments requires symmetry.

Since $P_s^*\bar{\mu}$ remains centered and 1-strongly log-concave, the de Bruijn argument as in [3, Corollary 5.9] applied with (5.39) yields

$$\mathrm{Ent}_\gamma(P_s^*\bar{\mu}) \le e^{-4s}\,\mathrm{Ent}_\gamma(\bar{\mu}). \tag{5.45}$$

The mean of $P_s^*\mu$ is $e^{-s}m$, and its centered translate is $P_s^*\bar{\mu}$. The Gaussian entropy translation formula therefore yields (5.41).

For the general case, set $\mu_\varepsilon = P_\varepsilon^*\bar{\mu}$. By Lemma 3.2, applied with parameter $e^{2\varepsilon} - 1$, these measures satisfy the required smooth two-sided Hessian bounds and converge to $\bar{\mu}$ in W_2. Moreover, the data processing inequality, lower semicontinuity, and the Ornstein–Uhlenbeck gradient commutation give

$$\mathrm{Ent}_\gamma(\mu_\varepsilon) \longrightarrow \mathrm{Ent}_\gamma(\bar{\mu}), \qquad \mathrm{I}_\gamma(\mu_\varepsilon) \le e^{-2\varepsilon}\mathrm{I}_\gamma(\bar{\mu}). \tag{5.46}$$

These facts allow passage to the limit in (5.38), (5.39) and (5.40). For (5.41), use $P_s^*\mu_\varepsilon = P_{s+\varepsilon}^*\bar{\mu}$ and the lower semicontinuity of entropy before letting $\varepsilon \to 0$. □

Remark 5.6. The two rates in (5.41) are optimal. Translates of γ attain the e^{-2s} term, while centered Gaussian contractions with covariance approaching I show that e^{-4s} cannot

be improved for centered measures. For a centered 1-strongly log-concave measure, the result of Bobkov, Gozlan, Roberto, and Samson [5, Corollary 2.2] gives

$$n\left(e^{2\,\mathrm{Ent}_\gamma(\mu)/n}-1\right)\le \mathrm{I}_\gamma(\mu).$$

Under the same centering and curvature assumptions, (5.39) doubles the left-hand side. This improvement uses the full geodesic Poincaré estimate.

A Failure of interpolation by the endpoint Poincaré constants

Proposition A.1. *There exist $\kappa>0$ and two κ-strongly log-concave probability measures μ_0,μ_1 on $\mathbb{R}^2$, such that their Wasserstein midpoint satisfies*

$$\sqrt{C_P(\mu_{1/2})}>\frac{1}{2}\sqrt{C_P(\mu_0)}+\frac{1}{2}\sqrt{C_P(\mu_1)}. \tag{A.1}$$

Proof. **Step 1: The square-shear counterexample.** Let

$$\Omega=\left(-\frac{\pi}{2},\frac{\pi}{2}\right)^2\subseteq\mathbb{R}^2,\qquad S=\begin{pmatrix}0&1\\1&0\end{pmatrix},$$
$$B_s=\begin{pmatrix}1&s\\s&1\end{pmatrix},\qquad \nu_s=(B_s)_{\#}m_\Omega.$$

where m_Ω is normalized Lebesgue measure on Ω. Since the restrictions of $C_c^\infty(\mathbb{R}^2)$ to Ω are dense in $H^1(\Omega)$, pulling the Rayleigh quotient back by B_s yields

$$C_P(\nu_s)^{-1}=\lambda(s):=\inf_{\substack{u\in H^1(\Omega)\setminus\{0\}\\ \int_\Omega u\,\mathrm{d}m_\Omega=0}}\frac{\int_\Omega\left\langle B_s^{-2}\nabla u,\nabla u\right\rangle\,\mathrm{d}m_\Omega}{\int_\Omega u^2\,\mathrm{d}m_\Omega}. \tag{A.2}$$

Note that $S^2=I$ and

$$B_s^{-2}=I-2sS+3s^2I+O(s^3),\qquad s\to0^+. \tag{A.3}$$

Consequently, for $u,v\in H^1(\Omega)$,

$$\begin{aligned}\int_\Omega\left\langle B_s^{-2}\nabla u,\nabla v\right\rangle\,\mathrm{d}m_\Omega=\int_\Omega\langle\nabla u,\nabla v\rangle\,\mathrm{d}m_\Omega-2s\int_\Omega\left(\partial_1u\,\partial_2v+\partial_2u\,\partial_1v\right)\mathrm{d}m_\Omega\\+3s^2\int_\Omega\langle\nabla u,\nabla v\rangle\,\mathrm{d}m_\Omega+O(s^3).\end{aligned} \tag{A.4}$$

The numerator in (A.2) defines a real-analytic family of closed symmetric forms, with common form domain $H^1(\Omega)$ and compact resolvent. We next determine the expansion of $\lambda(s)$ as $s\to0$. At $s=0$, separation of variables shows that the first positive Neumann eigenvalue on Ω is 1, with eigenspace

$$E=\mathrm{span}\{e_1,e_2\},\qquad e_1(x,y)=\sqrt{2}\sin x,\qquad e_2(x,y)=\sqrt{2}\sin y, \tag{A.5}$$

and the next eigenvalue is 2. In the orthonormal basis (e_1,e_2), the restriction to E of the first-order form in (A.4) has matrix $-16S/\pi^2$. Its two eigenvalues are distinct. The lower one is $-16/\pi^2$, with normalized eigenfunction $u_0=(e_1+e_2)/\sqrt{2}=\sin x+\sin y$.

Analytic perturbation theory for closed symmetric forms [31, Chapter VII, Section 4.6] therefore gives two analytic eigenvalue branches issuing from the double eigenvalue 1. Because their first derivatives at 0 are distinct, the lower branch is simple for all sufficiently small $s > 0$ and equals $\lambda(s)$. Thus

$$\lambda(s) = 1 - \frac{16}{\pi^2}s + \beta s^2 + O(s^3), \qquad s \to 0^+. \tag{A.6}$$

We claim that $\beta > \frac{192}{\pi^4}$. Indeed, let $I = (-\pi/2, \pi/2)$ and $\mathrm{d}m_I = \mathrm{d}x/\pi$, and consider the normalized Neumann basis

$$\phi_0 = 1, \qquad \phi_k(x) = \begin{cases} \sqrt{2}\sin(kx), & k \geq 1 \text{ odd}, \\ \sqrt{2}\cos(kx), & k \geq 2 \text{ even}. \end{cases} \tag{A.7}$$

Then $\phi_m \otimes \phi_n$ is an orthonormal Neumann eigenbasis on Ω, with eigenvalue $m^2 + n^2$. Set

$$A_m = \int_I \cos x \, \phi_m(x) \, \mathrm{d}m_I(x), \qquad D_m = \int_I \phi_m'(x) \, \mathrm{d}m_I(x).$$

Then

$$A_m^2 = \begin{cases} 4/\pi^2, & m = 0, \\ 8/\big(\pi^2(m^2-1)^2\big), & m \geq 2 \text{ even}, \\ 0, & m \text{ odd}, \end{cases} \qquad D_m^2 = \begin{cases} 8/\pi^2, & m \text{ odd}, \\ 0, & m \text{ even}. \end{cases} \tag{A.8}$$

Let q_1 and q_2 denote the coefficients of s and s^2 in (A.4). Then $q_2(u_0, u_0) = 3$, while

$$q_1(u_0, \phi_m \otimes \phi_n) = -2(A_m D_n + D_m A_n).$$

Because the lower eigenvalue of $q_1|_E$ is simple, the second-order formula for a split eigenvalue applies. Indeed, if $\alpha = -16/\pi^2$, then $q_1(u_0, v) = \alpha \langle u_0, v \rangle_{L^2(m_\Omega)}$ for every $v \in E$; hence the forcing term in the first-order eigenvector equation is orthogonal to E. The formula is therefore

$$\beta = 3 - \sum_{\substack{m,n \geq 0 \\ m^2+n^2 > 1}} \frac{4(A_m D_n + D_m A_n)^2}{m^2 + n^2 - 1}. \tag{A.9}$$

Using the parity relations above in (A.9), we obtain

$$\begin{aligned} \beta &= 3 - 8A_0^2 \sum_{\substack{n \geq 3 \\ n \text{ odd}}} \frac{D_n^2}{n^2 - 1} - 8 \sum_{\substack{m \geq 2 \text{ even} \\ n \geq 1 \text{ odd}}} \frac{A_m^2 D_n^2}{m^2 + n^2 - 1} \\ &= 3 - \frac{64}{\pi^4} - \frac{512}{\pi^4} D, \end{aligned} \tag{A.10}$$

where

$$\begin{aligned} D := \sum_{\substack{m \geq 2 \text{ even} \\ n \geq 1 \text{ odd}}} \frac{1}{(m^2-1)^2(m^2+n^2-1)} &= \frac{\pi}{4} \sum_{k=1}^{\infty} \frac{\tanh\left(\frac{\pi}{2}\sqrt{4k^2-1}\right)}{(4k^2-1)^{5/2}} \\ &< \frac{\pi}{4} \sum_{k=1}^{\infty} \frac{1}{3^{5/2} k^5} < \frac{1}{16}. \end{aligned}$$

Consequently,

$$\beta > 3 - \frac{96}{\pi^4} > \frac{192}{\pi^4}, \tag{A.11}$$

which proves the claim.

Therefore, by (A.2) and (A.6), for all sufficiently small $r > 0$,

$$\sqrt{C_P(\nu_{r/2})} - \left(\frac{1}{2}\sqrt{C_P(\nu_0)} + \frac{1}{2}\sqrt{C_P(\nu_r)}\right) = \left(\frac{\beta}{8} - \frac{24}{\pi^4}\right) r^2 + O(r^3) > 0. \tag{A.12}$$

Step 2: Strongly log-concave approximation. For $\delta > 0$, set

$$\mathrm{d}\eta_\delta(x) = Z_\delta^{-1} e^{-\delta|x|^2/2} \mathbf{1}_\Omega(x)\,\mathrm{d}x, \qquad \eta_{\delta,s} = (B_s)_\# \eta_\delta.$$

Here Z_δ is the normalizing constant. Up to an additive constant, the potential of $\eta_{\delta,s}$ is $V_{\delta,s}(y) = \frac{\delta}{2}|B_s^{-1}y|^2 + \iota_{B_s\bar{\Omega}}(y)$, where ι_K is zero on K and $+\infty$ on K^c. Since $\iota_{B_s\bar{\Omega}}$ is convex and

$$B_s^{-2} \succeq \frac{1}{(1+r)^2} I, \qquad 0 \le s \le r, \tag{A.13}$$

the measures $\eta_{\delta,s}$ are $\delta(1+r)^{-2}$-strongly log-concave. Moreover, $\frac{\mathrm{d}\eta_\delta}{\mathrm{d}m_\Omega} \in [e^{-\delta\pi^2/4}, e^{\delta\pi^2/4}]$. Hence, by (A.2),

$$e^{-\delta\pi^2/2} C_P(\nu_s) \le C_P(\eta_{\delta,s}) \le e^{\delta\pi^2/2} C_P(\nu_s), \qquad 0 \le s \le r, \tag{A.14}$$

which implies $C_P(\eta_{\delta,s}) \to C_P(\nu_s)$ uniformly on $[0, r]$. Thus, for $\delta > 0$ sufficiently small,

$$\sqrt{C_P(\eta_{\delta,r/2})} > \frac{1}{2}\sqrt{C_P(\eta_{\delta,0})} + \frac{1}{2}\sqrt{C_P(\eta_{\delta,r})}. \tag{A.15}$$

Fix such a $\delta > 0$. Since B_r is symmetric positive definite, it is the Brenier map from $\eta_{\delta,0}$ to $\eta_{\delta,r}$. Moreover, $(1-t)I + tB_r = B_{tr}$, so their Wasserstein geodesic is $(\eta_{\delta,tr})_{0\le t\le 1}$. Taking $\mu_0 = \eta_{\delta,0}$ and $\mu_1 = \eta_{\delta,r}$ gives

$$\sqrt{C_P(\mu_{1/2})} > \frac{1}{2}\sqrt{C_P(\mu_0)} + \frac{1}{2}\sqrt{C_P(\mu_1)}, \tag{A.16}$$

which proves the proposition. □

References

[1] M. Agueh and G. Carlier, *Barycenters in the Wasserstein space*, SIAM J. Math. Anal. **43** (2011), no. 2, 904–924, doi:10.1137/100805741.

[2] G. Aishwarya and D. Li, *Entropic and functional forms of the dimensional Brunn–Minkowski inequality in Gauss space*, Math. Ann. **393** (2025), no. 3–4, 3025–3042, doi:10.1007/s00208-025-03294-4.

[3] G. Aishwarya and L. Rotem, *New Brunn–Minkowski and functional inequalities via convexity of entropy*, Adv. Math. **490** (2026), 110841, doi:10.1016/j.aim.2026.110841.

[4] F. Bolley, I. Gentil, and A. Guillin, *Dimensional improvements of the logarithmic Sobolev, Talagrand and Brascamp–Lieb inequalities*, Ann. Probab. **46** (2018), no. 1, 261–301, doi:10.1214/17-AOP1184.

[5] S. G. Bobkov, N. Gozlan, C. Roberto, and P.-M. Samson, *Bounds on the deficit in the logarithmic Sobolev inequality*, J. Funct. Anal. **267** (2014), no. 11, 4110–4138, doi:10.1016/j.jfa.2014.09.016.

[6] H. J. Brascamp and E. H. Lieb, *On extensions of the Brunn–Minkowski and Prékopa–Leindler theorems, including inequalities for log concave functions, and with an application to the diffusion equation*, J. Funct. Anal. **22** (1976), 366–389.

[7] H. Brezis, *Functional Analysis, Sobolev Spaces and Partial Differential Equations*, Universitext, Springer, New York, 2011, doi:10.1007/978-0-387-70914-7.

[8] Y. Brenier, *Polar factorization and monotone rearrangement of vector-valued functions*, Comm. Pure Appl. Math. **44** (1991), 375–417.

[9] L. A. Caffarelli, *Monotonicity properties of optimal transportation and the FKG and related inequalities*, Comm. Math. Phys. **214** (2000), no. 3, 547–563; erratum, Comm. Math. Phys. **225** (2002), no. 2, 449–450, doi:10.1007/s002200000257.

[10] X. Cheng and D. Zhou, *Eigenvalues of the drifted Laplacian on complete metric measure spaces*, Commun. Contemp. Math. **19** (2017), no. 1, 1650001, 17 pp.

[11] D. Cordero-Erausquin and A. Eskenazis, *Concavity principles for weighted marginals*, arXiv:2506.16941v1 [math.FA] (2025).

[12] D. Cordero-Erausquin and A. Figalli, *Regularity of monotone transport maps between unbounded domains*, Discrete Contin. Dyn. Syst. **39** (2019), no. 12, 7101–7112, doi:10.3934/dcds.2019297.

[13] D. Cordero-Erausquin, R. J. McCann, and M. Schmuckenschläger, *A Riemannian interpolation inequality à la Borell, Brascamp and Lieb*, Invent. Math. **146** (2001), no. 2, 219–257, doi:10.1007/s002220100160.

[14] T. A. Courtade and M. Fathi, *Stability of the Bakry–Émery theorem on* $\mathbb{R}^n$, J. Funct. Anal. **279** (2020), no. 2, 108523, 28 pp., doi:10.1016/j.jfa.2020.108523.

[15] T. A. Courtade, M. Fathi, and A. Pananjady, *Existence of Stein kernels under a spectral gap, and discrepancy bounds*, Ann. Inst. Henri Poincaré Probab. Stat. **55** (2019), no. 2, 777–790, doi:10.1214/18-AIHP898.

[16] G. De Philippis and A. Figalli, *Rigidity and stability of Caffarelli's log-concave perturbation theorem*, Nonlinear Anal. **154** (2017), 59–70, doi:10.1016/j.na.2016.10.006.

[17] M. D. Donsker and S. R. S. Varadhan, *Asymptotic evaluation of certain Markov process expectations for large time. IV*, Comm. Pure Appl. Math. **36** (1983), no. 2, 183–212.

[18] A. Dytso, H. V. Poor, and S. Shamai (Shitz), *Conditional mean estimation in Gaussian noise: a meta derivative identity with applications*, IEEE Trans. Inform. Theory **69** (2023), no. 3, 1883–1898, doi:10.1109/TIT.2022.3216012.

[19] B. Efron, *Tweedie's formula and selection bias*, J. Amer. Statist. Assoc. **106** (2011), no. 496, 1602–1614, doi:10.1198/jasa.2011.tm11181.

[20] M. Erbar, K. Kuwada, and K.-T. Sturm, *On the equivalence of the entropic curvature-dimension condition and Bochner's inequality on metric measure spaces*, Invent. Math. **201** (2015), no. 3, 993–1071.

[21] A. Eskenazis, A. Giannopoulos, and N. Tziotziou, *Functional perimeter and the dimensional Brunn–Minkowski inequality for log-concave measures*, arXiv:2605.02747v2 [math.MG] (2026).

[22] A. Eskenazis and G. Moschidis, *The dimensional Brunn–Minkowski inequality in Gauss space*, J. Funct. Anal. **280** (2021), 108914.

[23] A. Eskenazis and Y. Shenfeld, *Intrinsic dimensional functional inequalities on model spaces*, J. Funct. Anal. **286** (2024), no. 7, Paper No. 110338, 56 pp.

[24] M. Fathi, *Stein kernels and moment maps*, Ann. Probab. **47** (2019), no. 4, 2172–2185, doi:10.1214/18-AOP1305.

[25] R. J. Gardner and A. Zvavitch, *Gaussian Brunn–Minkowski inequalities*, Trans. Amer. Math. Soc. **362** (2010), no. 10, 5333–5353, doi:10.1090/S0002-9947-2010-04891-3.

[26] B.-X. Han, *Measure rigidity of synthetic lower Ricci curvature bound on Riemannian manifolds*, Adv. Math. **373** (2020), 107327, 31 pp., doi:10.1016/j.aim.2020.107327.

[27] B.-X. Han and D.-Y. Liu, *Wasserstein barycenter convexity detects Hilbertian geometry*, arXiv:2606.28213v1 [math.MG] (2026).

[28] L. Hörmander, *On interior regularity of the solutions of partial differential equations*, Comm. Pure Appl. Math. **11** (1958), 197–218, doi:10.1002/cpa.3160110205.

[29] R. Jiang and P. Koskela, *Isoperimetric inequality from the Poisson equation via curvature*, Comm. Pure Appl. Math. **65** (2012), no. 8, 1145–1168, doi:10.1002/cpa.21405.

[30] R. Jiang, P. Koskela, and D. Yang, *Isoperimetric inequality via Lipschitz regularity of Cheeger-harmonic functions*, J. Math. Pures Appl. (9) **101** (2014), no. 5, 583–598, doi:10.1016/j.matpur.2013.07.002.

[31] T. Kato, *Perturbation theory for linear operators*, 2nd ed., Springer-Verlag, Berlin, 1976.

[32] B. Klartag, *Poincaré inequalities and moment maps*, Ann. Fac. Sci. Toulouse Math. (6) **22** (2013), no. 1, 1–41.

[33] B. B. Klartag and A. V. Kolesnikov, *Eigenvalue distribution of optimal transportation*, Anal. PDE **8** (2015), no. 1, 33–55, doi:10.2140/apde.2015.8.33.

[34] B. Klartag and E. Putterman, *Spectral monotonicity under Gaussian convolution*, Ann. Fac. Sci. Toulouse Math. (6) **32** (2023), no. 5, 939–967, doi:10.5802/afst.1759.

[35] A. V. Kolesnikov, *Mass transportation and contractions*, MIPT Proc. **2** (2010), no. 4, 90–99; in Russian, with English translation available as arXiv:1103.1479.

[36] A. V. Kolesnikov, *Hessian metrics, $CD(K,N)$-spaces, and optimal transportation of log-concave measures*, Discrete Contin. Dyn. Syst. **34** (2014), no. 4, 1511–1532.

[37] A. V. Kolesnikov and G. V. Livshyts, *On the Gardner–Zvavitch conjecture: symmetry in inequalities of Brunn–Minkowski type*, Adv. Math. **384** (2021), 107689.

[38] A. V. Kolesnikov and E. Milman, *Riemannian metrics on convex sets with applications to Poincaré and log-Sobolev inequalities*, Calc. Var. Partial Differential Equations **55** (2016), no. 4, Art. 77, doi:10.1007/s00526-016-1018-3.

[39] A. V. Kolesnikov and E. Milman, *Poincaré and Brunn–Minkowski inequalities on the boundary of weighted Riemannian manifolds*, Amer. J. Math. **140** (2018), no. 5, 1147–1185.

[40] A. Malliaris, J. Melbourne, C. Roberto, and M. Roysdon, *Functional liftings of restricted geometric inequalities*, arXiv:2508.15247v3 [math.FA] (2025).

[41] R. J. McCann, *A convexity principle for interacting gases*, Adv. Math. **128** (1997), no. 1, 153–179, doi:10.1006/aima.1997.1634.

[42] G. Mijoule, M. Raič, G. Reinert, and Y. Swan, *Stein's density method for multivariate continuous distributions*, Electron. J. Probab. **28** (2023), Paper No. 59, 40 pp., doi:10.1214/22-EJP883.

[43] P. Nayar and T. Tkocz, *A note on a Brunn–Minkowski inequality for the Gaussian measure*, Proc. Amer. Math. Soc. **141** (2013), no. 11, 4027–4030, doi:10.1090/S0002-9939-2013-11609-6.

[44] F. Otto and C. Villani, *Generalization of an inequality by Talagrand and links with the logarithmic Sobolev inequality*, J. Funct. Anal. **173** (2000), no. 2, 361–400, doi:10.1006/jfan.1999.3557.

[45] F. Santambrogio and X.-J. Wang, *Convexity of the support of the displacement interpolation: counterexamples*, Appl. Math. Lett. **58** (2016), 152–158, doi:10.1016/j.aml.2016.02.016.

[46] A. Saumard and J. A. Wellner, *Log-concavity and strong log-concavity: a review*, Stat. Surv. **8** (2014), 45–114, doi:10.1214/14-SS107.

[47] J. Serres, *Behavior of the Poincaré constant along the Polchinski renormalization flow*, Commun. Contemp. Math. **26** (2024), no. 7, 2350035, doi:10.1142/S0219199723500359.

[48] C. Villani, *Optimal Transport: Old and New*, Grundlehren der mathematischen Wissenschaften, vol. 338, Springer-Verlag, Berlin, 2009, doi:10.1007/978-3-540-71050-9.

[49] M.-K. von Renesse and K.-T. Sturm, *Transport inequalities, gradient estimates, entropy and Ricci curvature*, Comm. Pure Appl. Math. **58** (2005), no. 7, 923–940, doi:10.1002/cpa.20060.

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